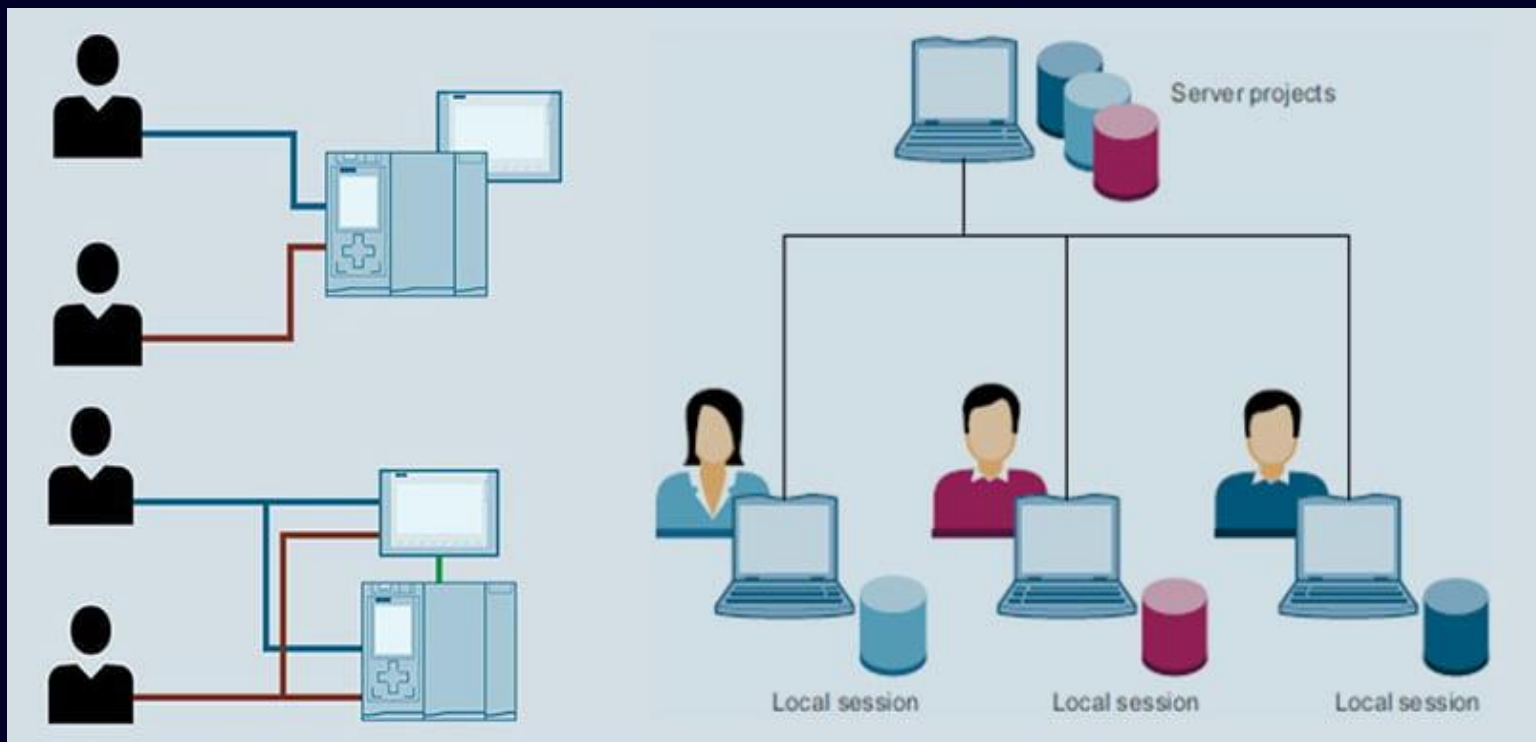


TIA Portal Tech Bytes

Multuser Engineering



SIEMENS

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Overview Of Document

This document aims to provide a short guide on configuring and using Multiuser Engineering within the TIA Portal framework. This guide shows how multiple people can work on the same automation project at the same time using a project server — the project server can be either a dedicated computer or a individuals computer, and it runs the server software and stores all the shared projects.

With the help of the TIA Project-Server you can:

- Create new server projects that multiple users can work on simultaneously.
- View and manage existing server projects.
- Create and manage local sessions.
- Roll back of changes.
- Track and comment on changes to projects.

Document Revision Information

REVISION INFORMATION			
Revision Number	Date	Author	Description
1	February 2026	Marcus Riley	Initial Release

Required Software Packages

This demonstration requires the following software packages to be installed. It is assumed that this is done prior to continuing to the engineering sections of this manual.

1. **TIA Project Server V2.2 – Installed On Computer That Will Host The TIA Project Server**
 - a. No Licence is required to run TIA Project Server
 - b. Download of this software available from
<https://support.industry.siemens.com/cs/document/109810588/tia-project-server?dti=0&lc=en-WW>
2. **TIA Portal V21 – STEP 7 + WinCC – Installed On The Computers That Will Access The Projects From TIA Project Server**
 - a. Including licence STEP 7 Professional V19 + WinCC Unified Comfort V19
 - b. A trial licence can be activated for 21 days
 - c. Download of this software available from
<https://support.industry.siemens.com/cs/document/109989774/simatic-step-7-incl-safety-s7-plcsim-and-wincc-v21-trial-download?dti=0&lc=en-AU>
3. **TIA Portal Multiuser Engineering V21– Installed On The Computers That Will Access The Projects From TIA Project Server**
 - a. Including licence TIA Portal Multiuser Engineering V21
 - b. A trial licence can be activated for 21 days
 - c. Installation of this option is available when installing the above TIA Portal.

Virtual Machine Availability

A virtual machine for training purposes is available upon request.

Important Note – Saving TIA Portal Project

It must be noted that TIA Portal V19, does not have a an “**Auto-Save**” feature. This is because if bulk engineering of a project is done and auto save was a feature it would not be possible to go back. If you wish to use “Auto-Save” then follow the below link to one created by a TIA Portal user via TIA openness.

- <https://github.com/Schievel1/TIAAutoSaver>

Basic System Architecture

Put basic system architecture here.

Important Note - User Management

When using TIA Project Server, user access and rights are based on what current users are configured in the Windows system that is hosting the TIA Project Server. It is important that these users are created before the creation of the security certificates as these certificates use the current Windows users. If the user is created after the certificate is created, then the certificate must be regenerated.

To configure new windows users, use this Microsoft guide - <https://support.microsoft.com/en-au/windows/manage-user-accounts-in-windows-104dc19f-6430-4b49-6a2b-e4dbd1dcdf32>

There are three pre-defined user groups as follows:

- **Manager** – Gives members of the group **Full Access** to the projects.
- **Contributor** – Gives members of the group **Partial Access (Cannot Delete)** to the projects.
- **Member** – Gives members of the group **Read Only Access** to the projects.

Important Note – Remote Access To TIA Project Server

When using TIA Project Server, it is not recommended to use this remotely. Should it be necessary to use TIA Project Server remotely then it must be done so using a VPN connection. This guide uses a VPN connection to connect remotely to a TIA Project Server located in Sydney Australia and is done so using a VPN connection.

Important Note – TIA Portal – TIA Project Server Licencing

When using TIA Project Server, depending on the functionality you are wanting to use it for, see below examples:

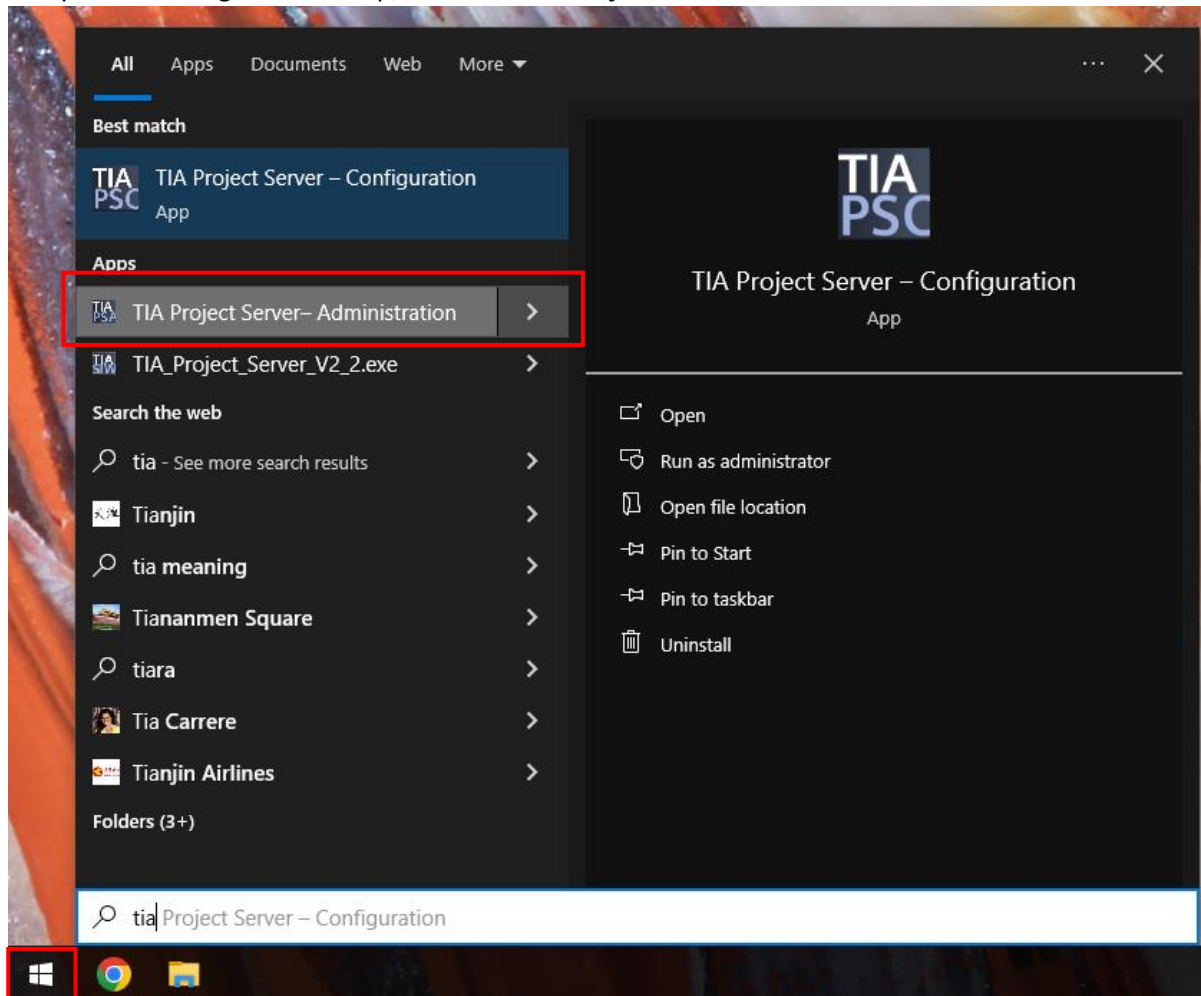
1. **Exclusive Engineering** – The only licences required are those that are directly connected to the project that you are working on.
 - a. For example: STEP 7 Professional is required when working on an S7-1500 PLC
 - i. This is regardless of accessing the project via TIA Project Server or not.
2. **Multiuser Engineering** – This is where more than one user wishes to access a project at the same time.
 - a. This requires a licence on each engineering PC that wants to access the project via the TIA Project Server.

ADD LICENCE PART NUMBERS -

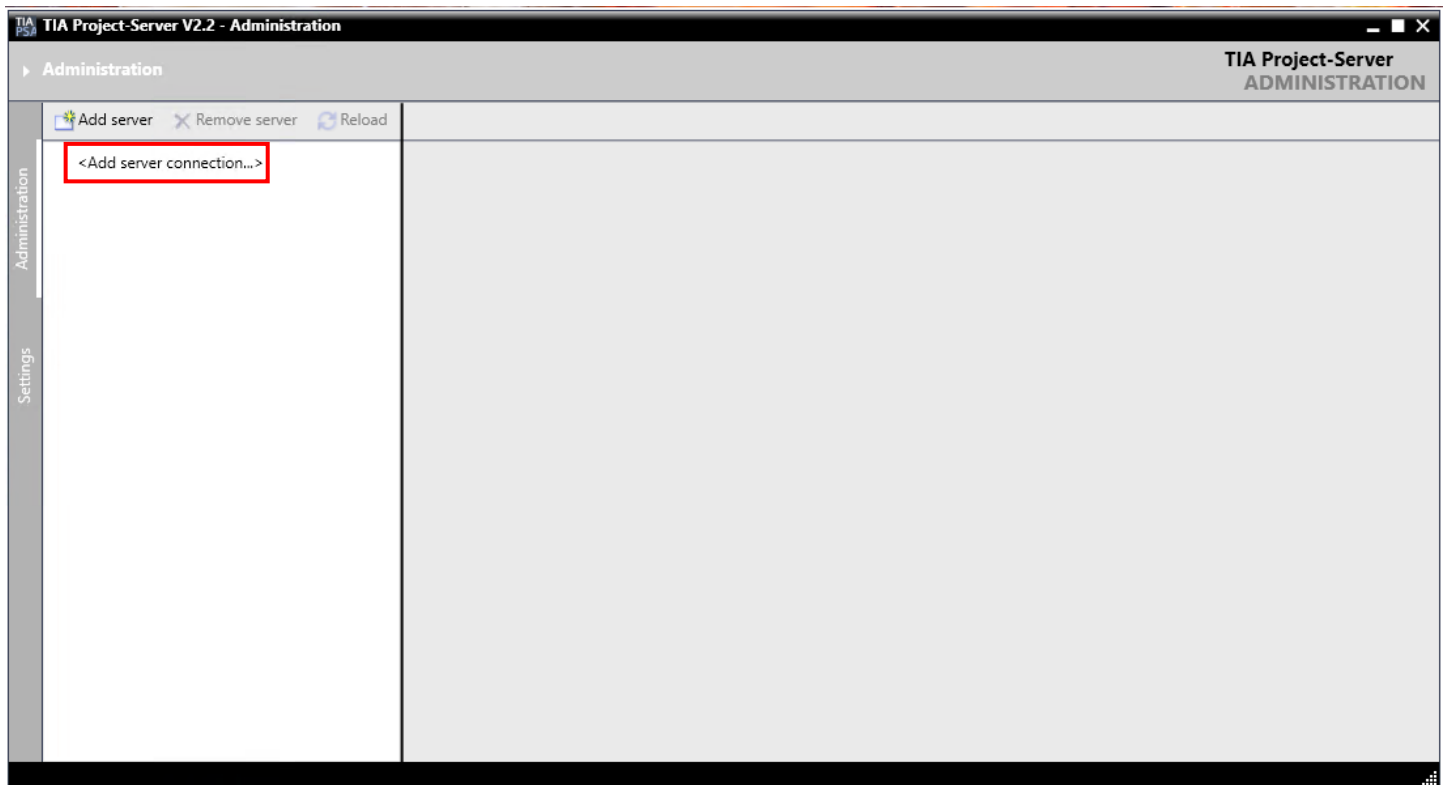
Configuring Users On TIA Project Server

To access the project server projects, the TIA Project Server must have registered users with various access rights. To do this, follow the below steps on the TIA Project Server machine, as shown in the basic system architecture section of this guide:

1. Click the **“Windows Start Button”**.
2. Type **“TIA Project Server - Administration”**. This will search for the TIA Project Server application.
3. To open the configuration setup, click the **“TIA Project Server – Administration”** icon.



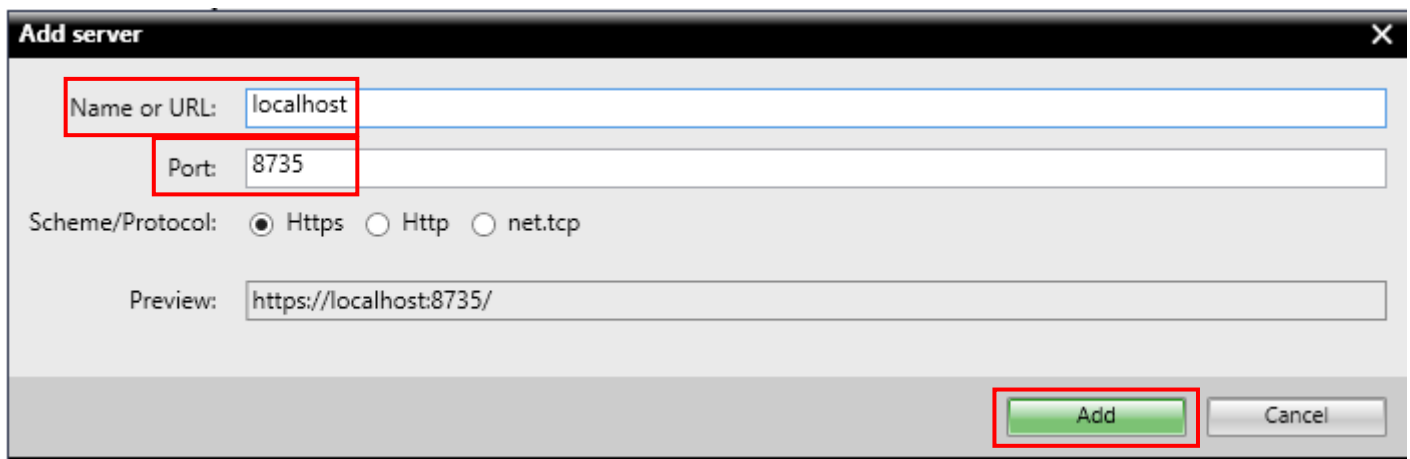
- The **"TIA Project Server – Administration"** program will open, double click on the **"Add Server Connection"** text to add a new server to the configuration.



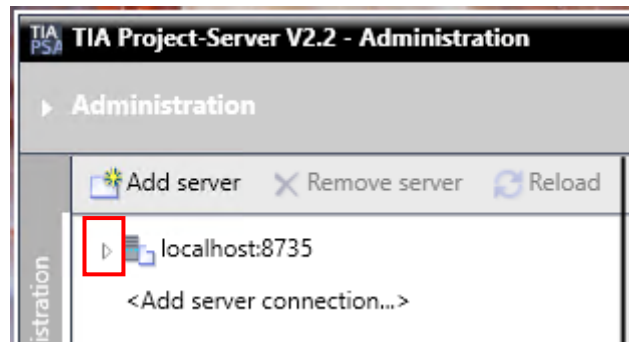
- The **"Add Server"** dialog box will appear, in the **"Name or URL"** entry box type **"localhost"**.

**** The server "localhost" is used in this as the TIA Project Server is located on the same machine ****

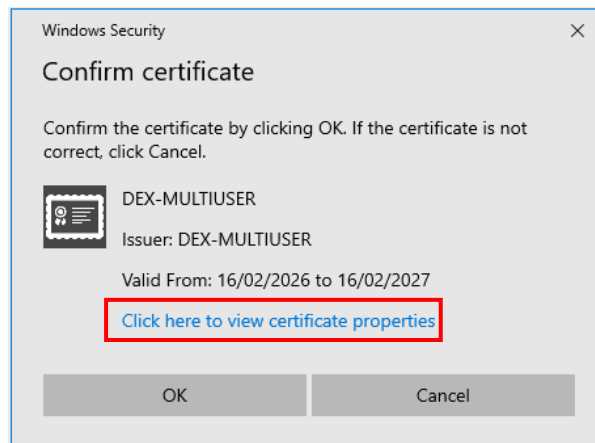
- In the **"Port"** entry box, enter the previously configured port, for this guide use **"8735"**.
- Click **"Add"** to add the server to the list of available servers.



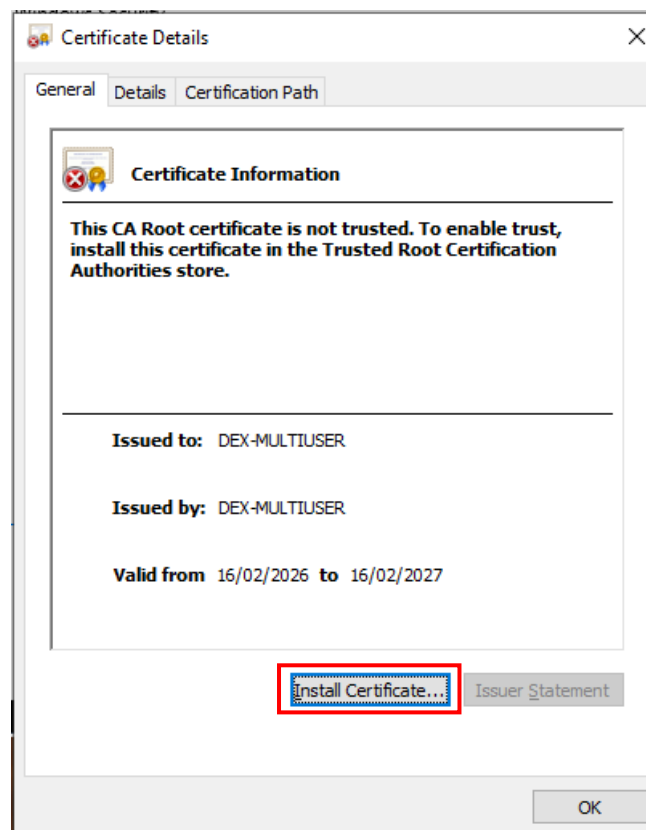
- Expand the **“localhost:8735”** server by using the right pointing arrow next to the text.



- A **“Confirm Certificate”** dialog will appear, click the **“Click here to view certificate properties”** button to accept and install the certificate.

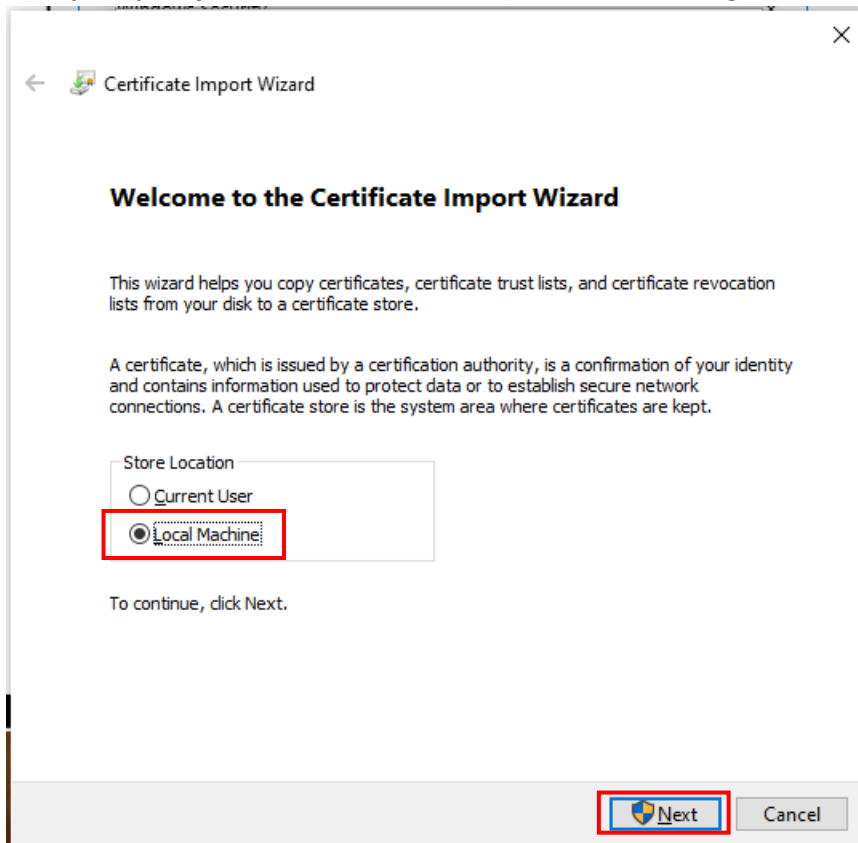


- The **“Certificate Details”** dialog box will appear, click the **“Install Certificate...”** button so that the certificate will be installed permanently on the TIA Project Server. This will stop this prompt from appearing each time the **“Administration”** section is accessed.

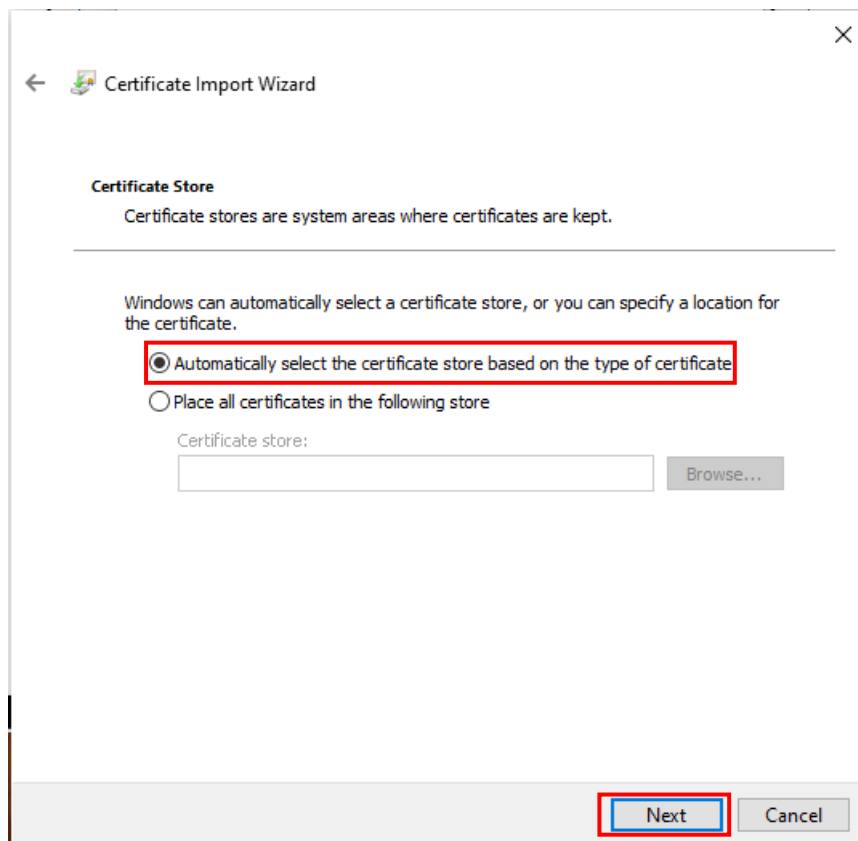


11. When the **"Certificate Import Wizard"** appears change the radio button to the **"Local Machine"** selection and click **"Next"**.

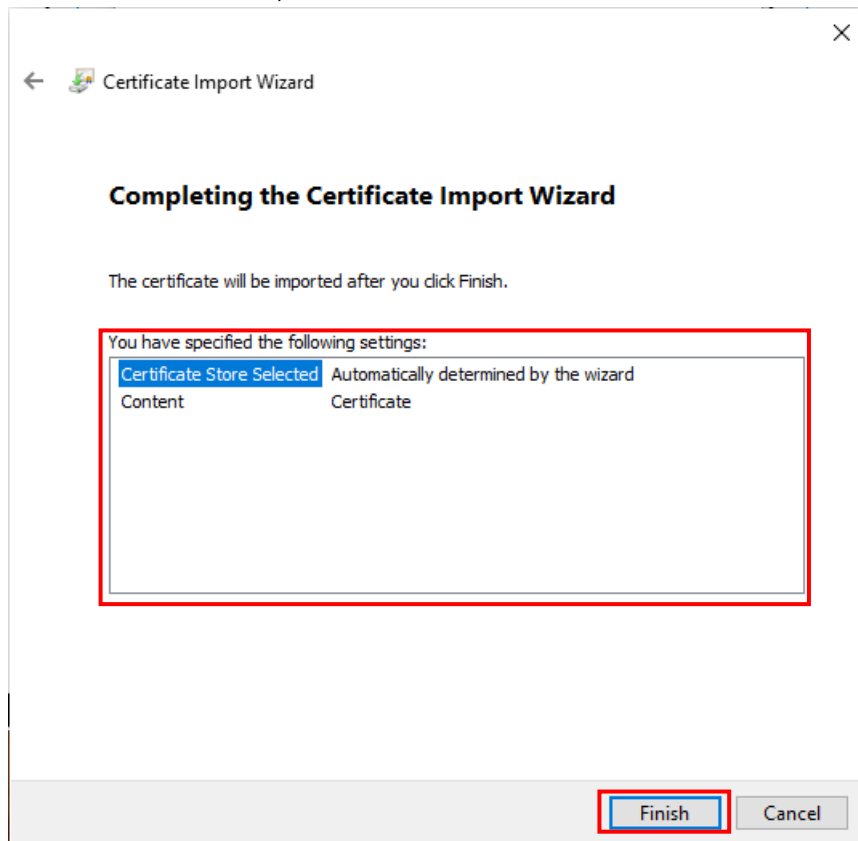
****You might be prompted for User Account Control (Administration Rights) to be able to progress.****



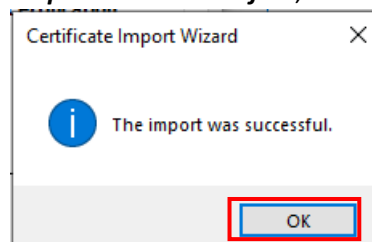
12. Ensure that the **"Automatically select the certificate store based on the type of certificate"** radio button is selected and click **"Next"**.



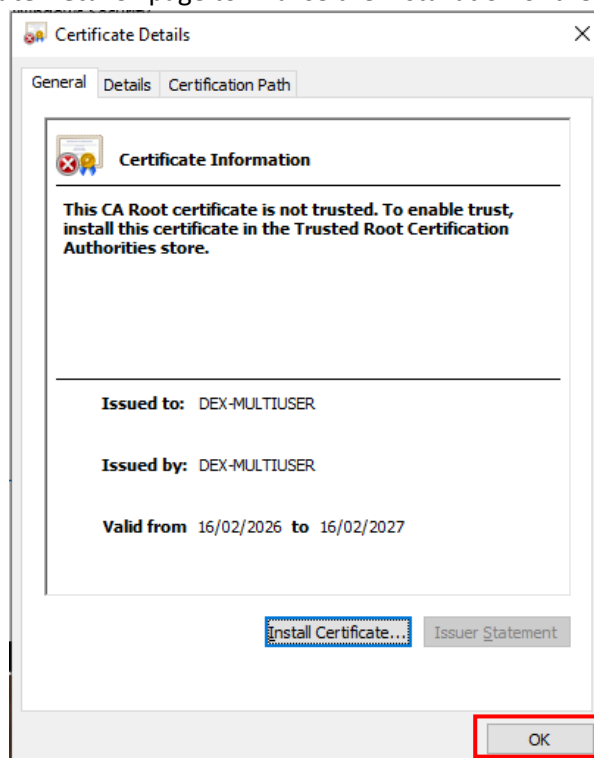
13. A summary of selections will be shown, double check these and click **“Finish”** to finalise the settings.



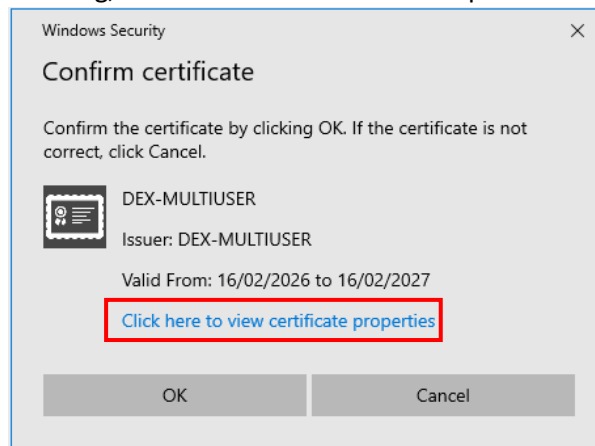
14. A prompt should appear saying **“The import was successful”**, click **“OK”** to continue.



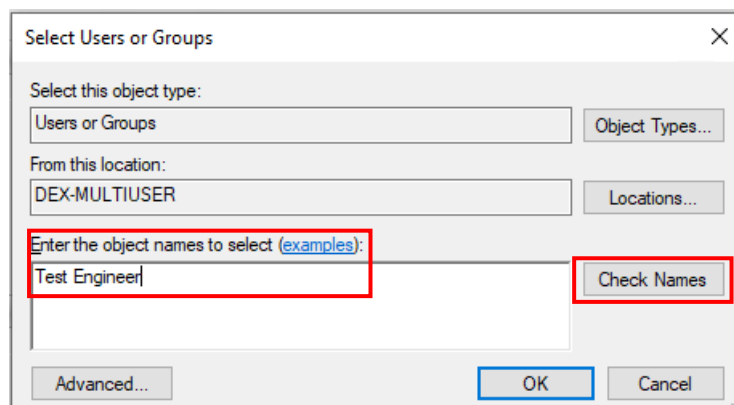
15. Click **“OK”** on the **“Certificate Details”** page to finalise the installation of the certificate.



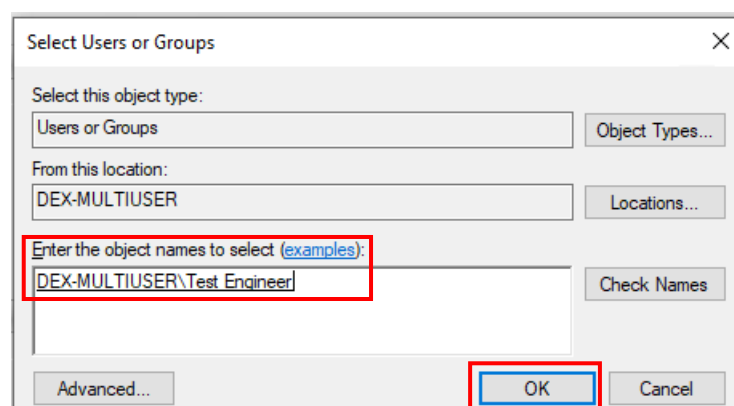
16. On the **“Confirm Certificate”** dialog, click the **“OK”** button to accept the certificate.



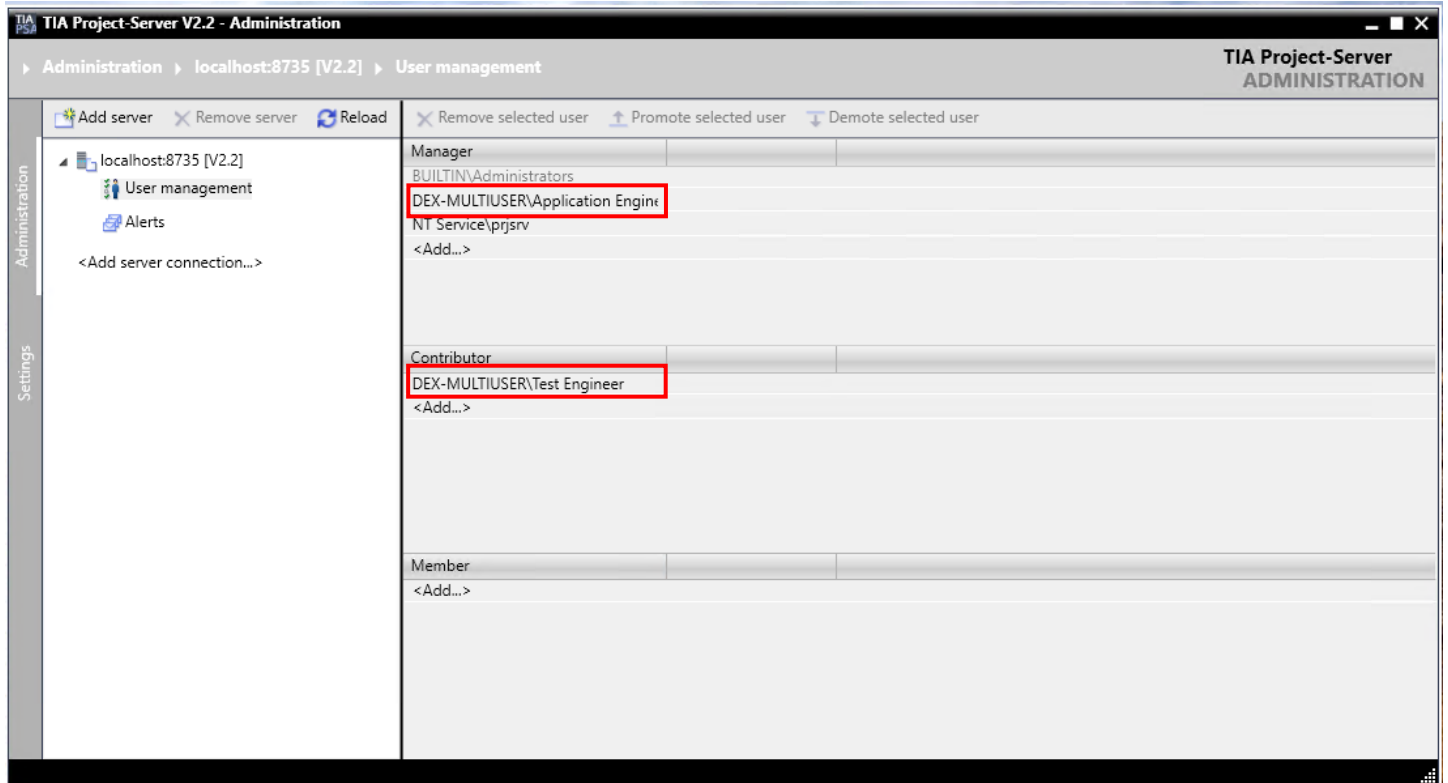
17. Under the **“User Management”** tab, you will be able to add users and setup their access rights. Users are based on what users are configured in the Windows system that is hosting the TIA Project Server.
- To configure new windows users, use this Microsoft guide - <https://support.microsoft.com/en-au/windows/manage-user-accounts-in-windows-104dc19f-6430-4b49-6a2b-e4dbd1dcdf32>
 - The Windows user must be created before the security certificate is created, see previous section.
 - The following users have been configured for this guide:
 - DEX-MULTIUSER/Application Engineer**
 - DEX-MULTIUSER/Test Engineer**
18. In the Manager section, you will see that the locally logged in user will be already setup. In this guide, **“Application Engineer”** is already part of this group.
19. In the **“Contributor”** section, double click the **“Add”** text.
20. The **“Select Users or Groups”** dialog will appear, in the **“Enter the object names to select”** text input, add the user you wish to add as a **“Contributor”**. For this example, we will use **“Test Engineer”** as this has already been configured on the TIA Project Server.
21. Click the **“Check Names”** button as this will search for the entered user and give it the full digital name.



22. Once the complete name has been found, it will be updated in the text box and its possible to click **“OK”** to accept the changes.



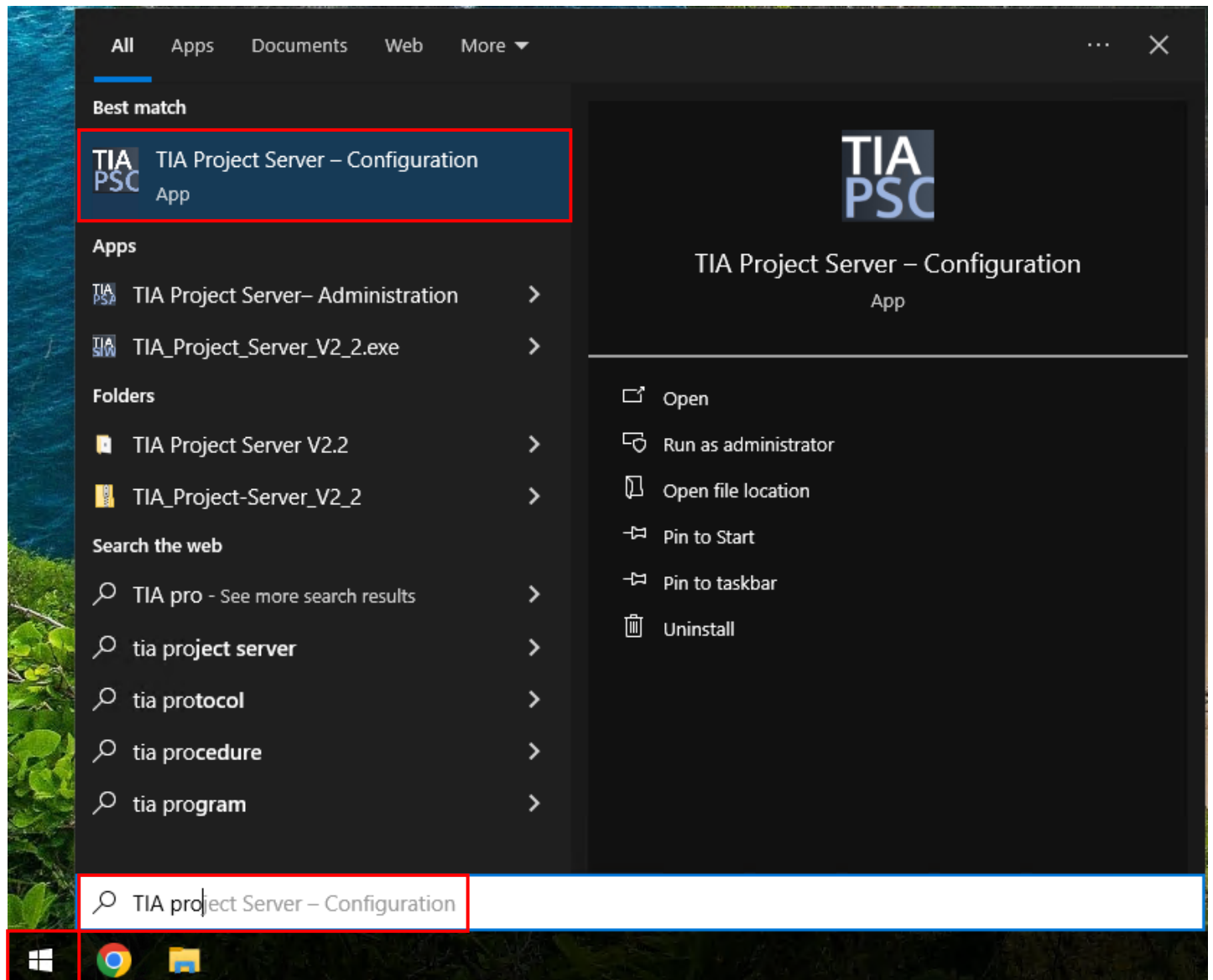
23. A user for the “**Manager**” group has been added (Application Engineer), and a user for the “**Contributor**” group has been added (Test Engineer), this completes this section.



Configuring TIA Project Server

Before using the TIA Project Server, some settings need to be configured so that the engineers TIA Portal machine can communicate with the TIA Project Server. To configure these settings follow the below steps, these must be completed on the computer that is running the TIA Project Server whether it is a dedicated computer or an individual's computer, as shown in the basic system architecture section of this guide:

1. Click the **"Windows Start Button"**.
2. Type **"TIA Project Server - Configuration"**. This will search for the TIA Project Server application.
3. To open the configuration setup, click the **"TIA Project Server – Configuration"** icon.



4. The **"TIA Project Server Configuration"** dialog box will appear. Under the **"Connection Settings"** heading, we will need to set the specific protocol to use. By default, this is set to **"https"** which is the most secure method. For this guide we will leave this setting as default, Siemens will always set the most secure method by default under its **"Security By Default"** policy.
5. The port for the connection must also be set, by default this is set to **"8735"** and it is recommended that this is left as default unless another program/service requires this port.
6. As this is the first time configuring the TIA Project server, you will need to **"Create a new self-signed certificate"** for the connection. Leave this radio button selected.
7. Under the **"Storage"** heading you will need to select the **"Storage Location For Server Projects"** this is a suitable folder location for all your TIA Portal Projects to be stored. For this guide will use the following folder located on the servers desktop – **C:\Users\Application Engineer\Desktop\TIA Portal Projects**. Use the "Browse" button to navigate to the appropriate folder.
8. Within this dialog box it is also possible to set the number of **"previous revisions"** to keep. By default it is set to **"10"**. We will leave it set to this number for the purpose of the guide.
9. Once all the above is configured, select the **"Install Service"** button to save and install the configuration.

TIA Project-Server V2.2 - Configuration

Server status

Client info: No data

Service status: Not installed Uninstall service

Server status: Stopped Start service

Connection settings

Protocol: ☐ http ☒ https (Recommended)

Configure Microsoft Windows so that you are only using secure SSL/TLS protocols and codes to achieve the highest level of security. See: <https://docs.microsoft.com/en-us/windows-server/identity/ad-fs/operations/manage-ssl-protocols-in-ad-fs>

Port: 8735 (default: 8735)

Timeout profile: Fast (default)

Security

☒ Create a new self-signed certificate.

☐ Use existing certificate

N. Subject Valid until

Storage

Storage location for server projects: C:\Users\Application Engineer\Desktop\TIA Portal Projects Browse...

Number of Saved revisions: 10

Install service Exit

English

10. Once the service has been installed, its possible to use the **“View Log”** link to understand any warnings/errors that might have occurred during the installation process.
11. Use the **“Start Service”** button to make the service available to any clients that wish to connect. This will take some time but once complete you will see a **“Action Successful Completed”** message at the bottom of the screen. You should also see the status of the server change from **“Stopped”** to **“Started”**.
12. You can use the **“Exit”** button to close the dialog process as the installation process is now complete.

TIA Project-Server V2.2 - Configuration

Server status

Client info:

Service status:

Server status:

Connection settings

Protocol: ☐ http ☒ https (Recommended)

Configure Microsoft Windows so that you are only using secure SSL/TLS protocols and codes to achieve the highest level of security. See: <https://docs.microsoft.com/en-us/windows-server/identity/ad-fs/operations/manage-ssl-protocols-in-ad-fs>

Port: (default: 8735)

Timeout profile:

Security

☐ Create a new self-signed certificate.

☒ Use existing certificate

Name	Subject	Valid until
Siemens Automation TIA Portal Project Server	CN=DEX-MULTIUSER	2/16/2027 8:43 AM

Storage

Storage location for server projects:

Number of Saved revisions:

Action successfully completed
[View log](#)

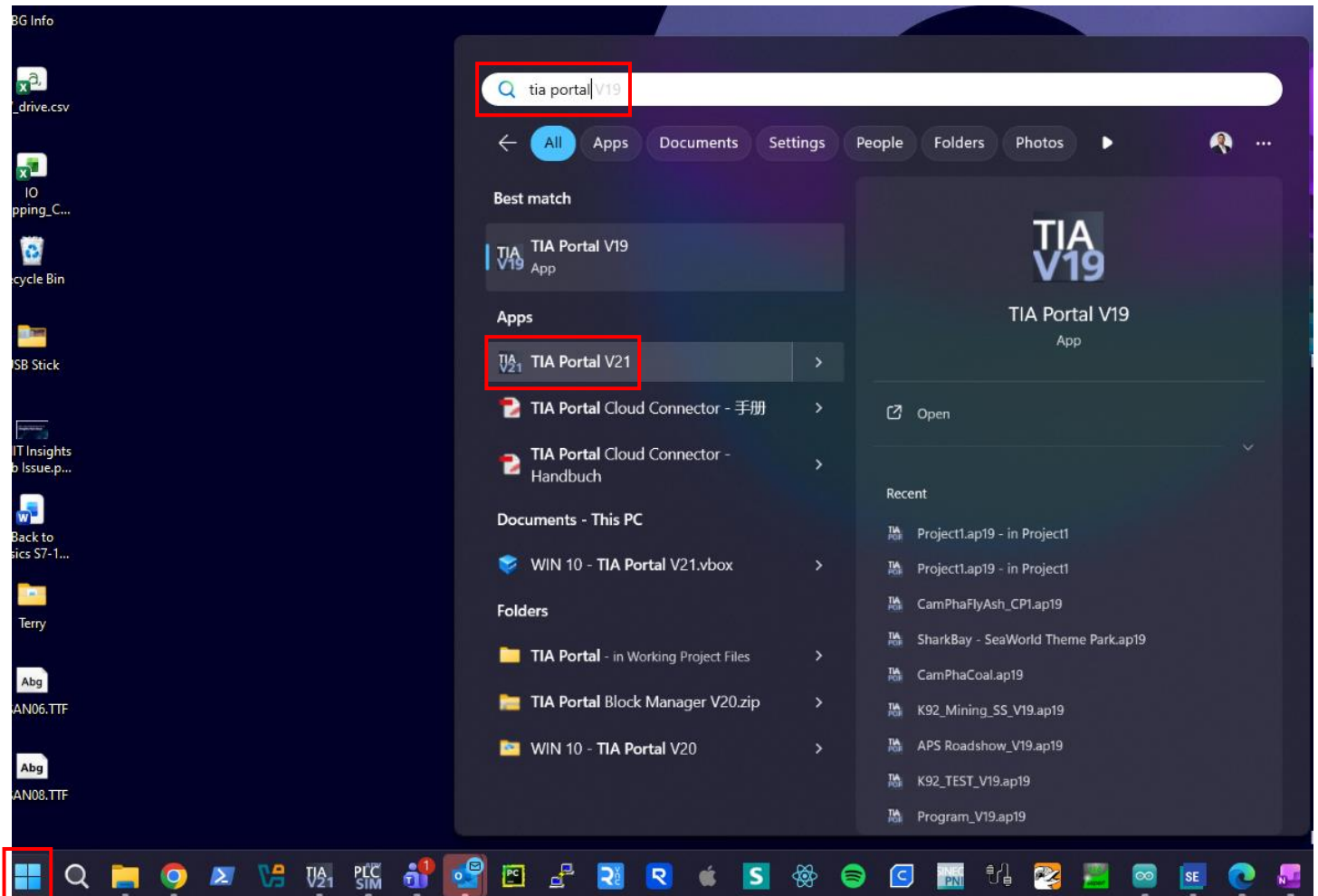
English

Configuring TIA Portal Project Server Connection On Client

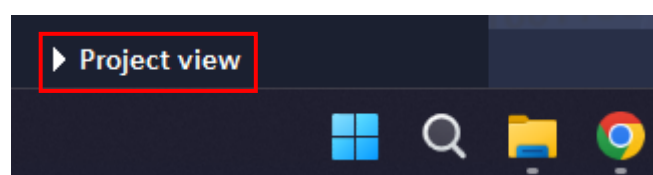
To access the project server projects, TIA Portal must be configured. To do this, follow the below steps on each of the client machines, as shown in the basic system architecture section of this guide:

1. Click the **“Windows Start Button”**.
2. Type **“TIA Portal”**. This will search for the TIA Portal application you wish to configure to communicate with the TIA Project Server.
3. To open **“TIA Portal”**, click the appropriate icon.

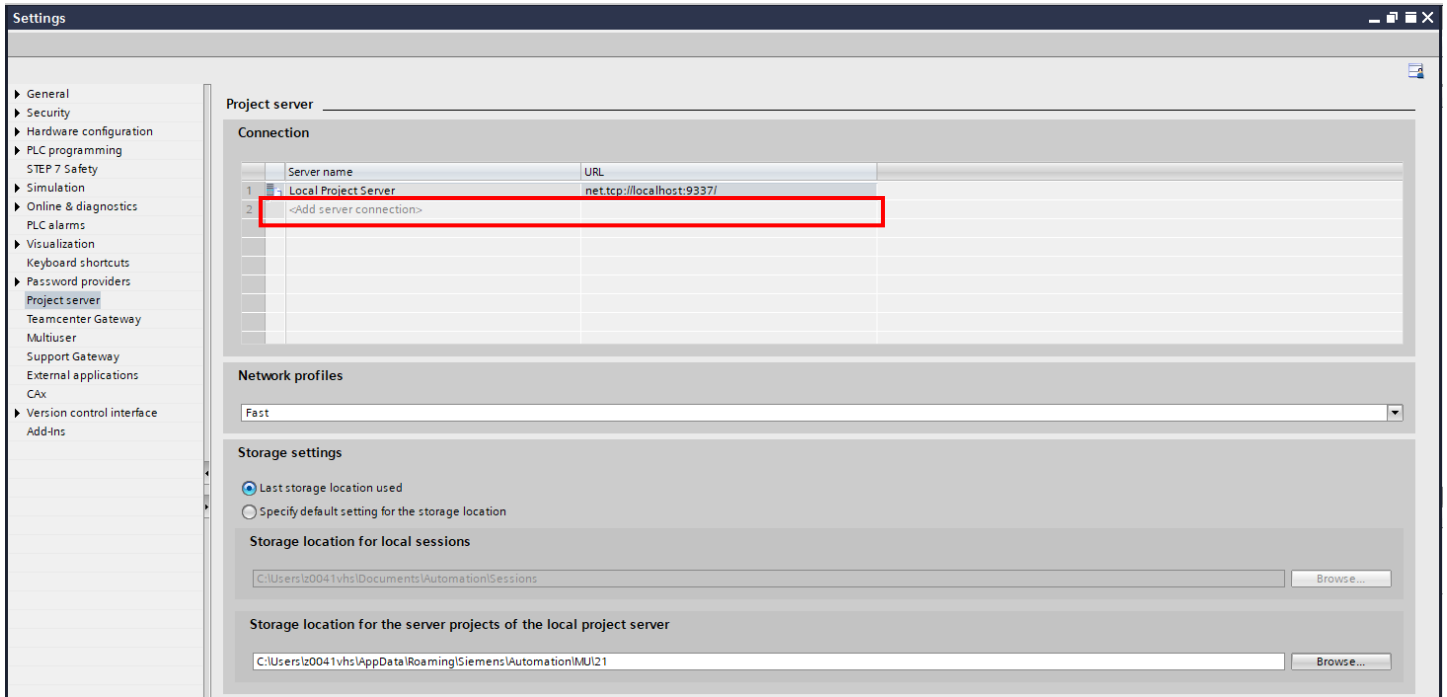
**** Its important to note that the TIA Project Server and TIA Portal work independent of each other, meaning you can have TIA Portal V19 connecting to TIA Project Server V2.2. The only thing you need to ensure is that you have the appropriate TIA Portal version for the project that you wish to open****



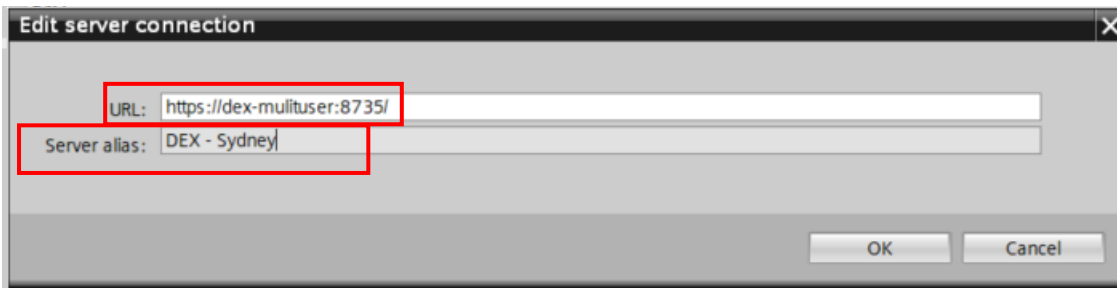
4. This will open **“TIA Portal”**, by default TIA Portal opens in **“Portal View”**. To configure the connection to the TIA Project Server you will need to change to **“Project View”**. To do this click the **“Project View”** link in the bottom left-hand corner of the TIA Portal window. If you are already in **“Project View”** you can skip this step.



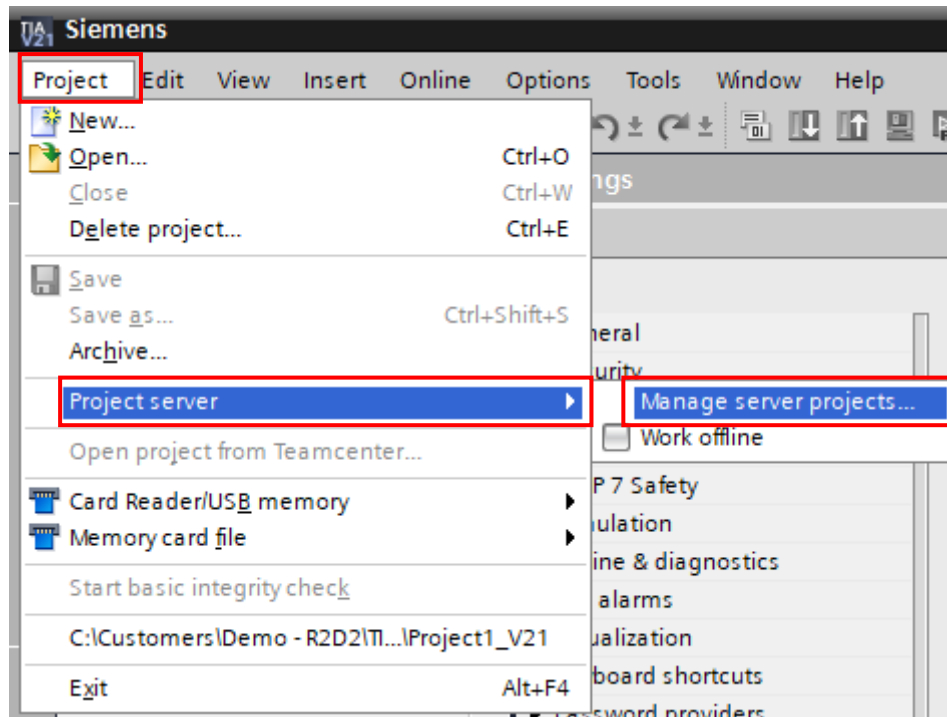
- You will be navigated to the **“Settings → Project Server”** page and a default server will be entered. Double click **“Add Server Connection”** to edit the settings.



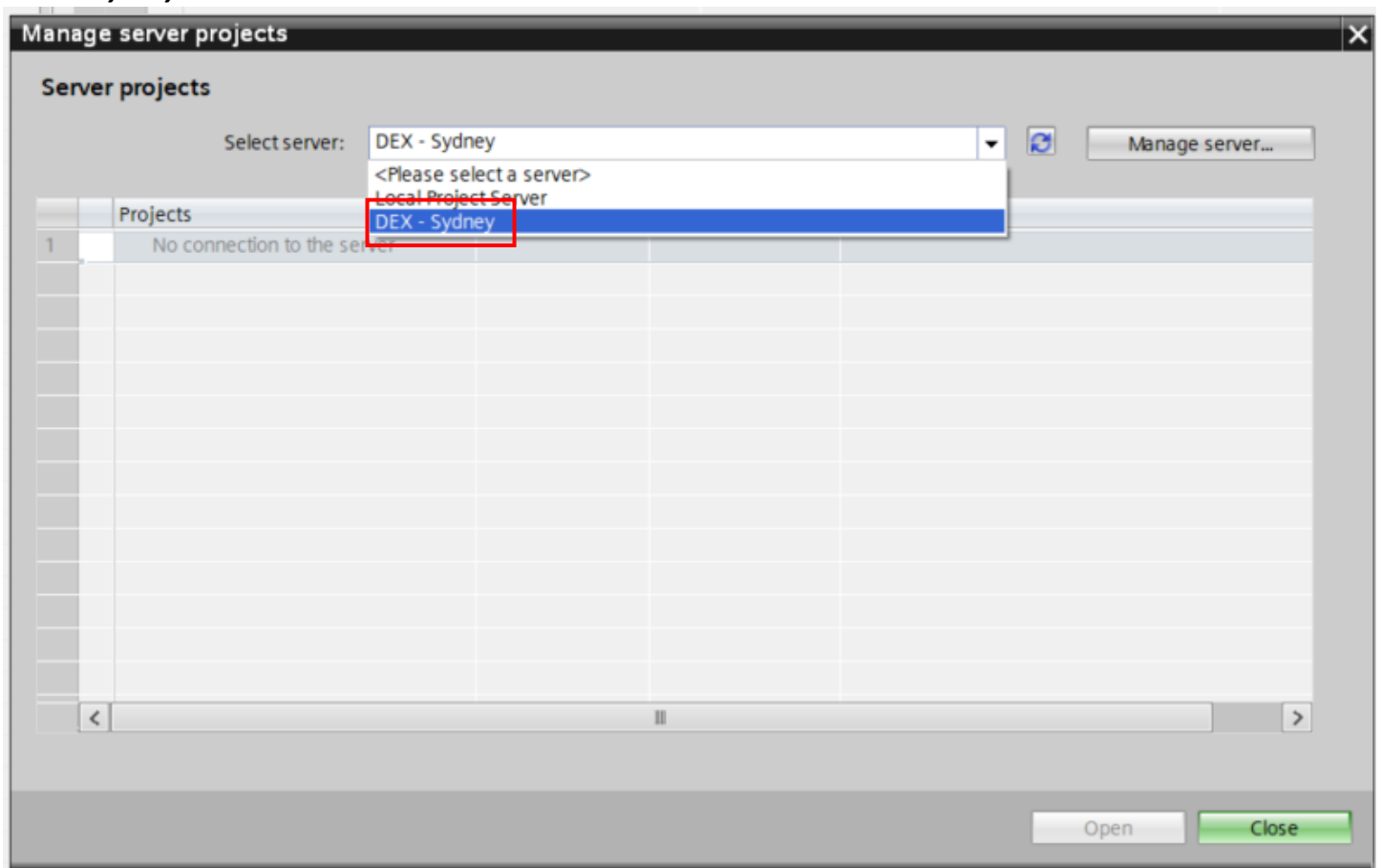
- The **“Add New Project Server Connection”** dialog box will appear, in the **“URL”** enter the URL of the TIA Project Server. This is usually an **“IP Address”** with the configured **“Port”**, a QDN (Qualified Domain Name) can also be used. For this guide we will use the configured TIA Project Server located in Siemens Limited Sydney which uses the following QDN – ***https://dex-multiuser:8735***
- Give the TIA Project Server a **“Server Alias”**, this is a name that is more familiar to the engineer so that it can be easily recognised later. For this guide, we have used **“DEX – Sydney”**.



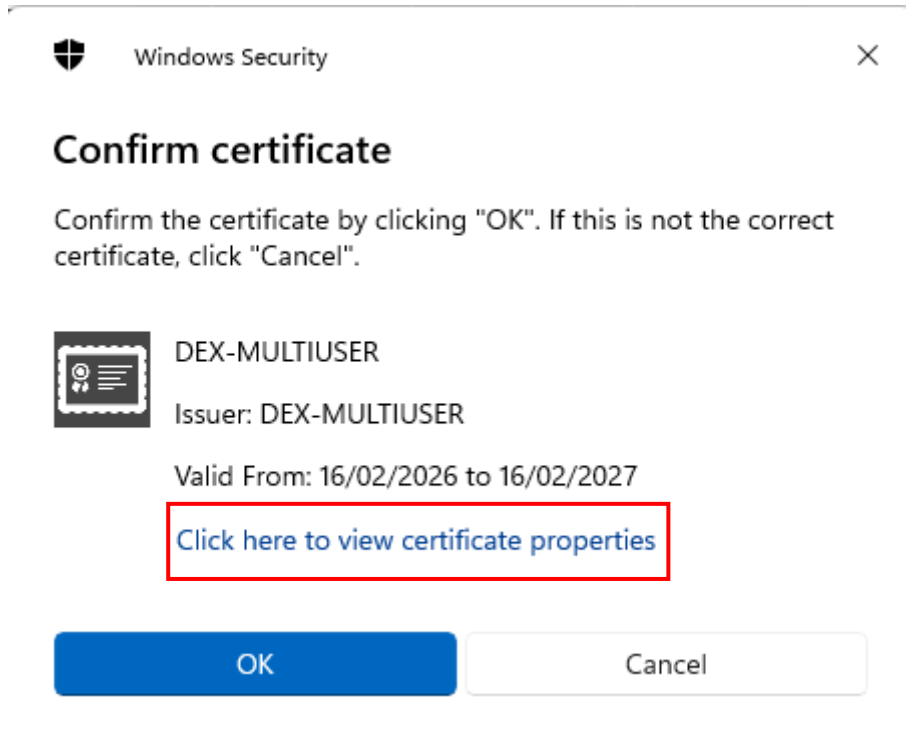
10. With the server now setup in TIA Portal, navigate back to **“Project → Project Server → Manage Server Projects”**.



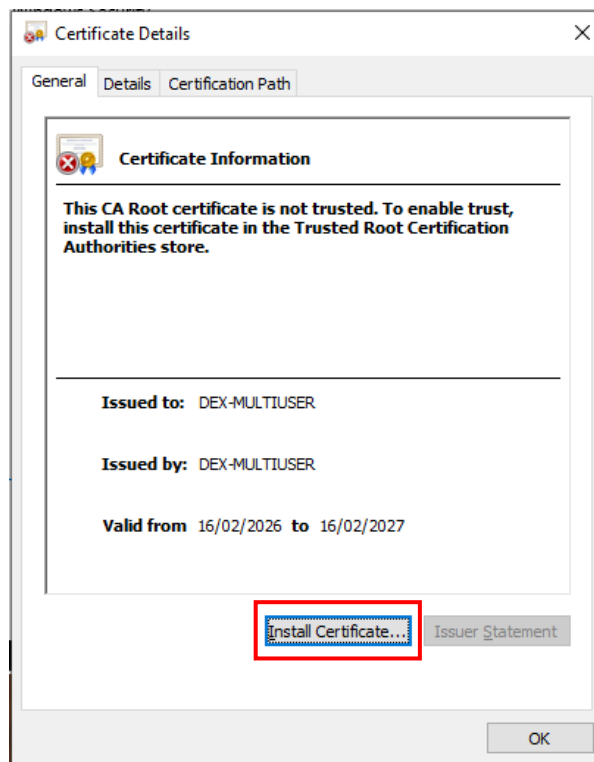
11. The **“Manage Server Projects”** dialog box will appear, from the **“Select Server”** dropdown box select the **“DEX - Sydney”** as the active server.



12. A **“Confirm Certificate”** prompt will appear. Click **“Click here to view certificate properties”** to start the prompts to install the certificate from the TIA Project Server.

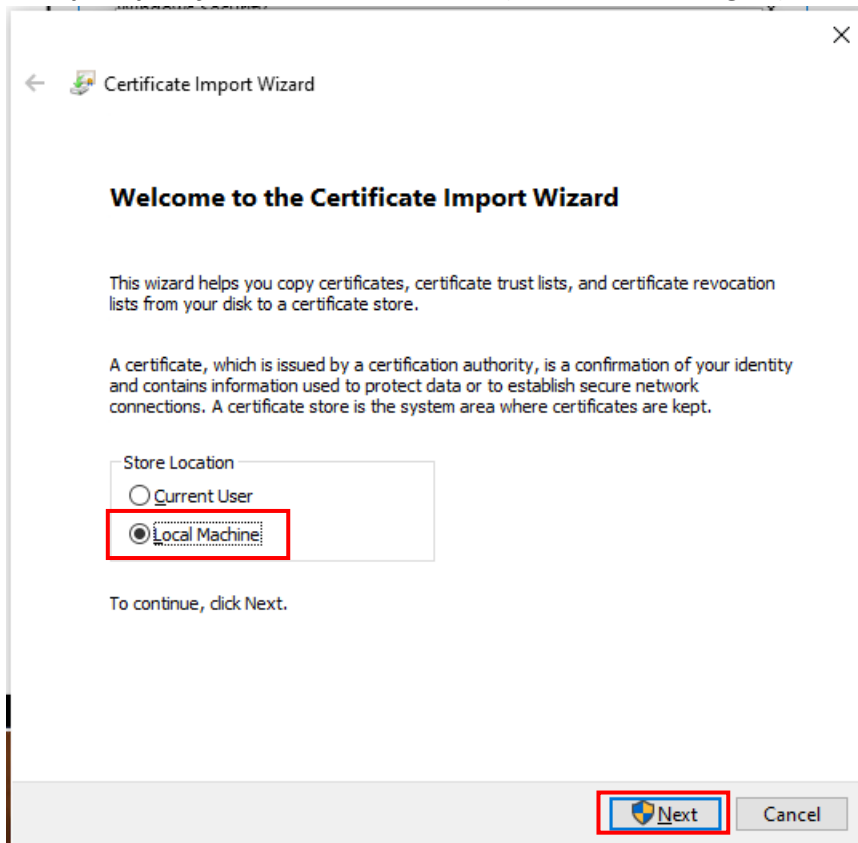


13. The **"Certificate Details"** dialog box will appear, click the **"Install Certificate..."** button to begin the installation of the certificate to the local machine.

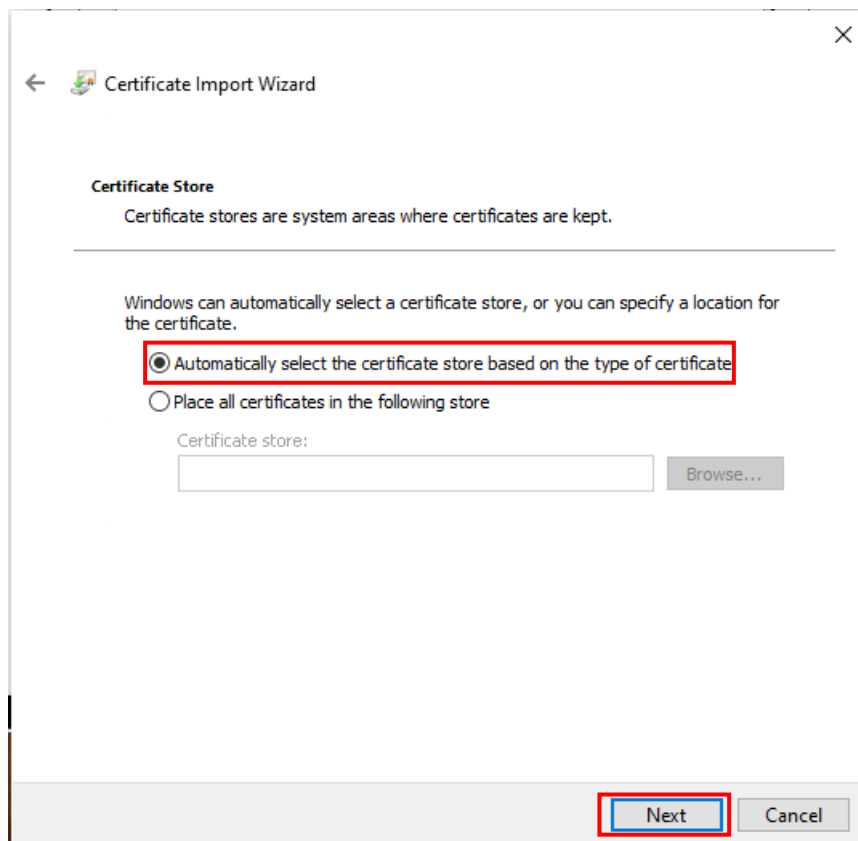


14. When the **"Certificate Import Wizard"** appears change the radio button to the **"Local Machine"** selection and click **"Next"**.

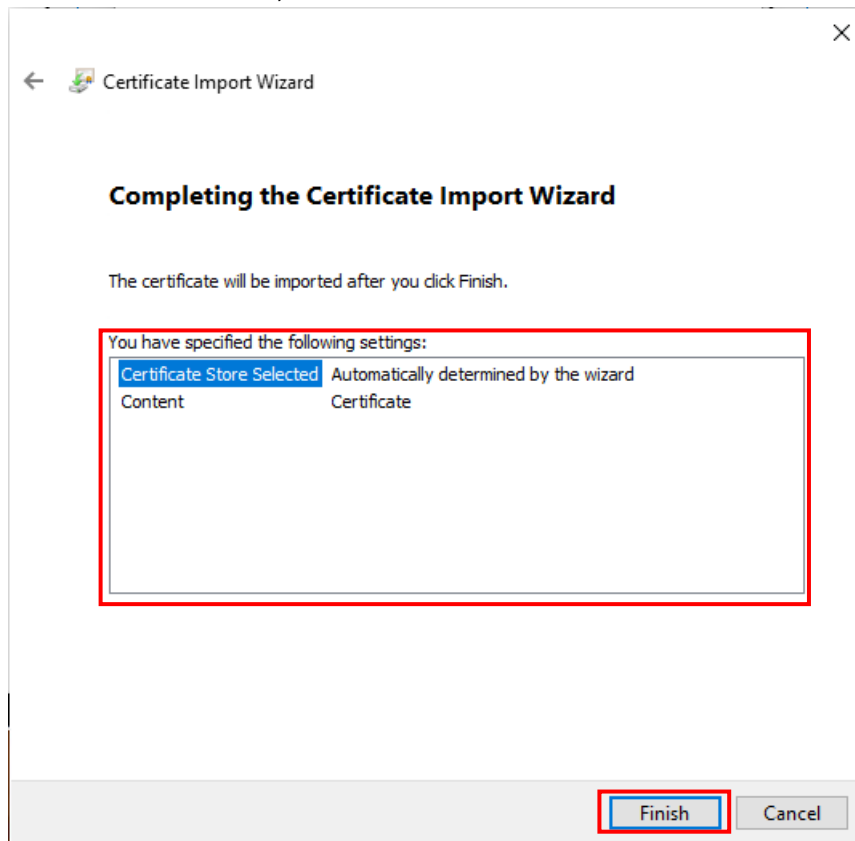
****You might be prompted for User Account Control (Administration Rights) to be able to progress.****



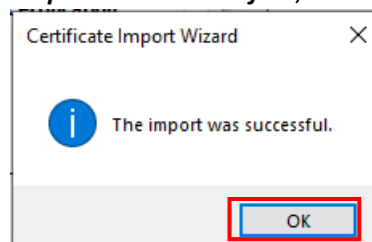
15. Ensure that the **"Automatically select the certificate store based on the type of certificate"** radio button is selected and click **"Next"**.



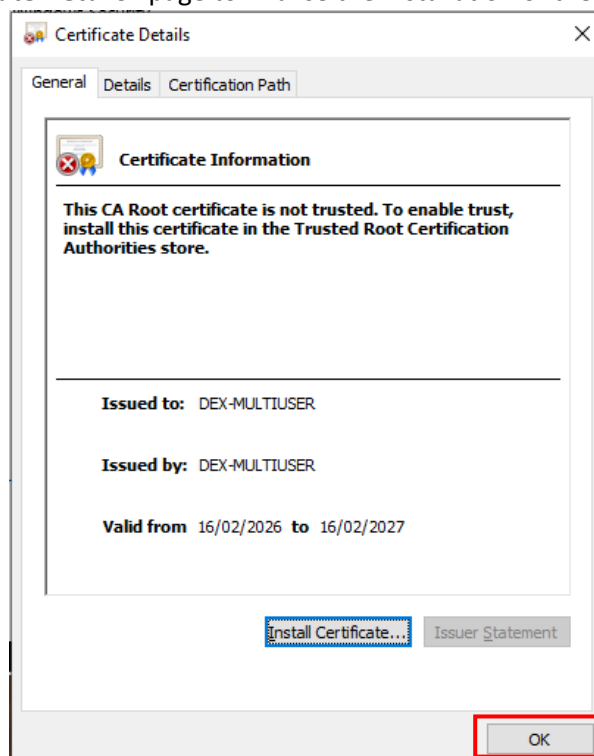
16. A summary of selections will be shown, double check these and click **“Finish”** to finalise the settings.



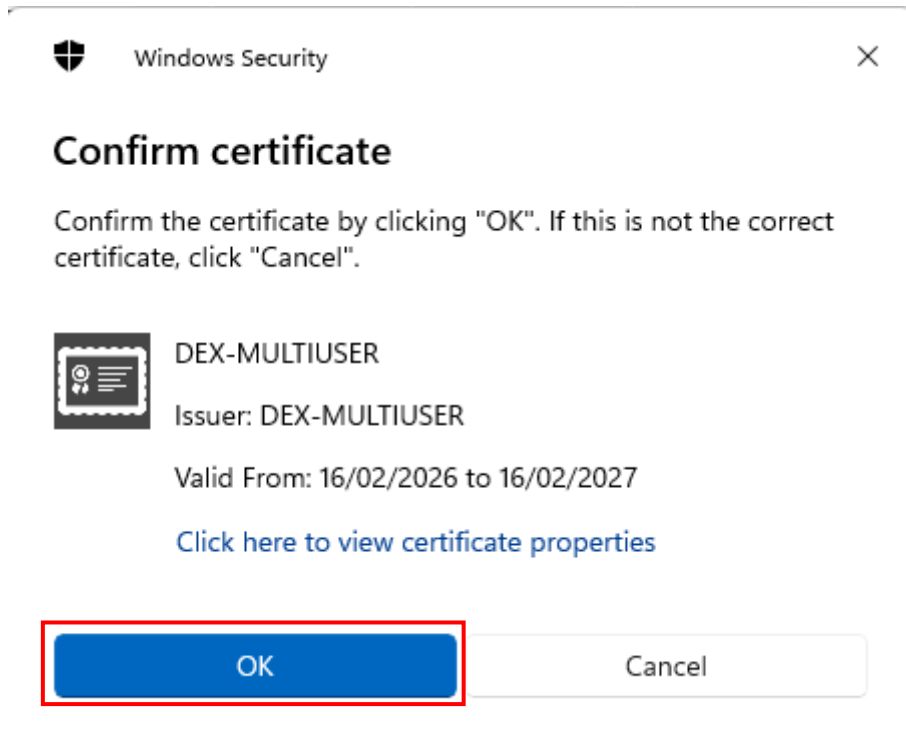
17. A prompt should appear saying **“The import was successful”**, click **“OK”** to continue.



18. Click **“OK”** on the **“Certificate Details”** page to finalise the installation of the certificate.



19. On the **“Confirm Certificate”** dialog, click the **“OK”** button to accept the certificate.

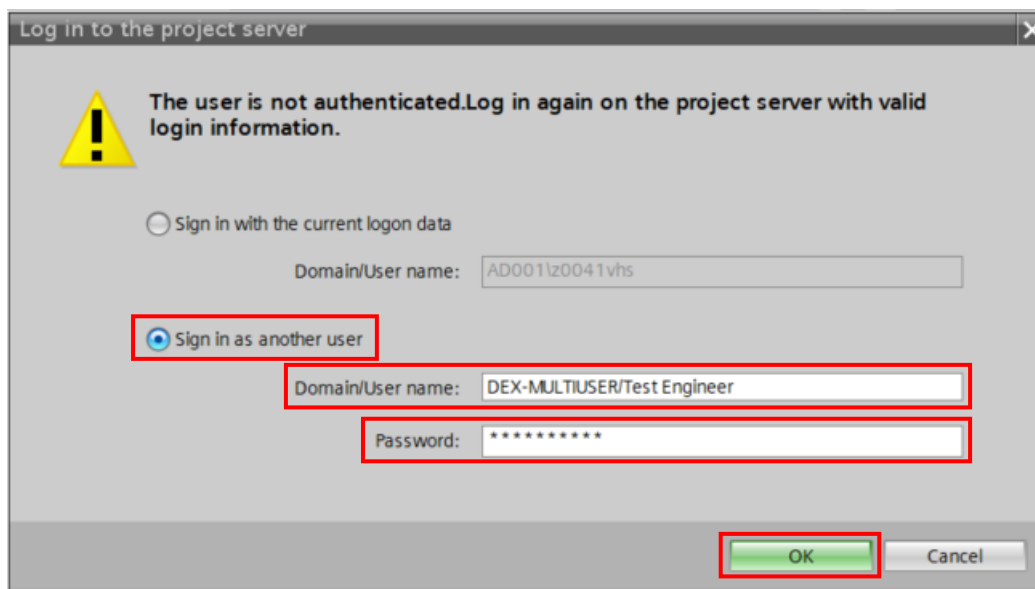


20. You will now be reverted to TIA Portal, the **“Log in to the project server”** dialog box will appear. Select the **“Sign in as another user”** radio button.

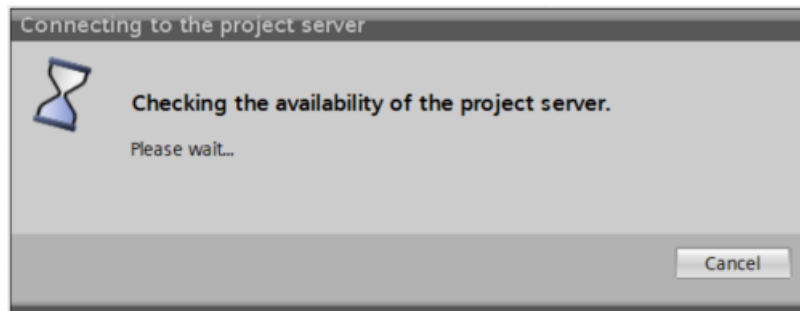
21. Enter the **“Domain/Username”** that was setup earlier, in the example we will use **“DEX-MULTIUSER/Test Engineer”**

22. Enter the **“Password”** that was setup earlier, in this example we will use **“Siemens123”**.

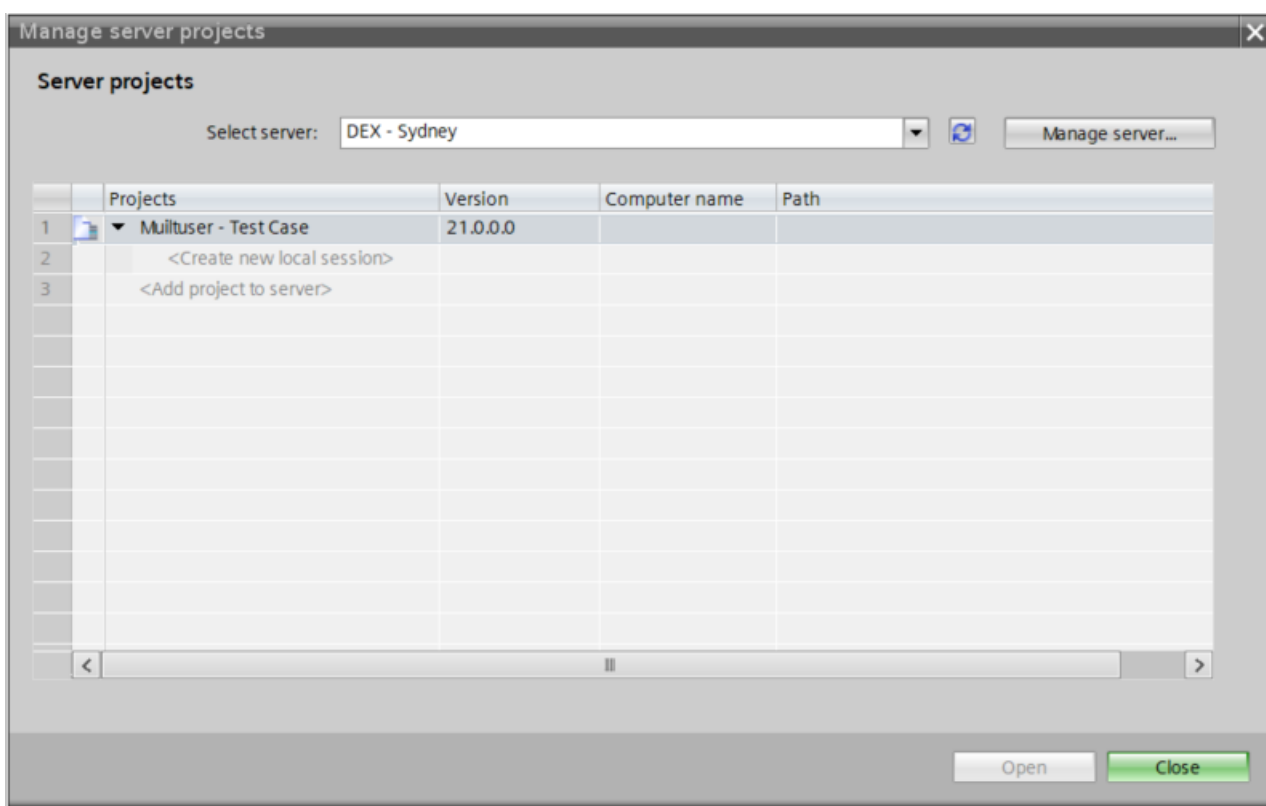
23. Click **“OK”** to log in to the TIA Project Server.



24. Enter the **“Password”** that was setup earlier, in this example we will use **“Siemens123”**.
25. Click **“OK”** to log in to the TIA Project Server. This will take a few seconds to connect and whilst it is connecting you will get the below.



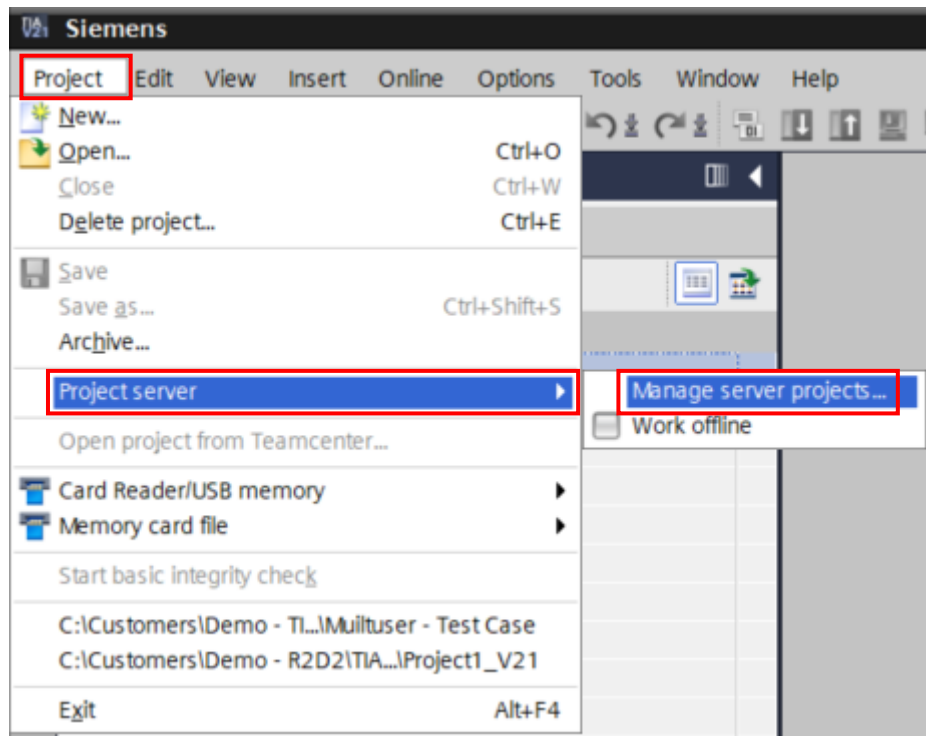
26. Once connected, you will see the projects that exist on the TIA Project Server. Below shows the currently hosted projects on the **DEX – Sydney TIA Project Server**.



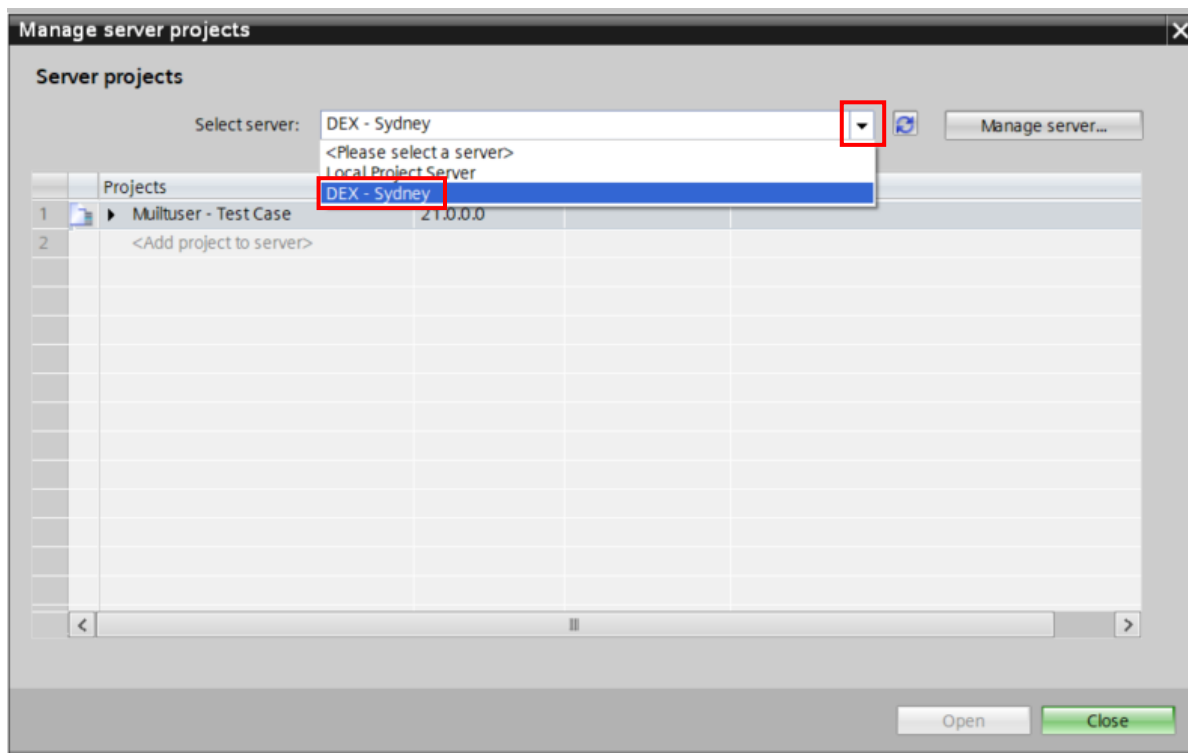
Creating A Local Session Of A TIA Project Server Hosted Project

When a user wants to co-create on a project that is hosted by the TIA Project Server they must create a local session of that project. To do this follow the below steps:

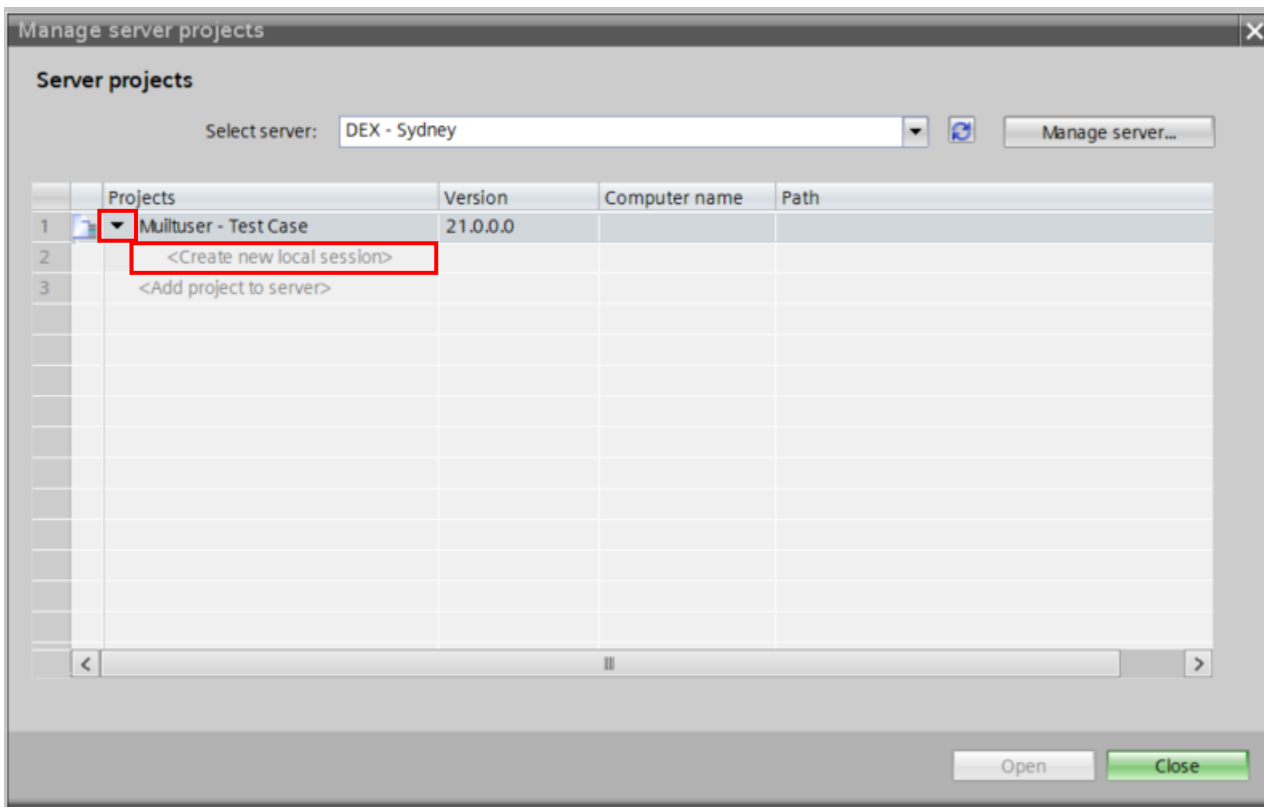
1. In TIA Portal, navigate to **"Project → Project Server → Manage Server Projects"**.



2. From the dropdown menu, select the server that you wish to connect to. In this guide, we will connect to the previously configured **"DEX – Sydney"** server. If you have previously logged in during this session you won't need to do that again. If you have not previously logged in please see previous chapters in this guide on the login procedure.

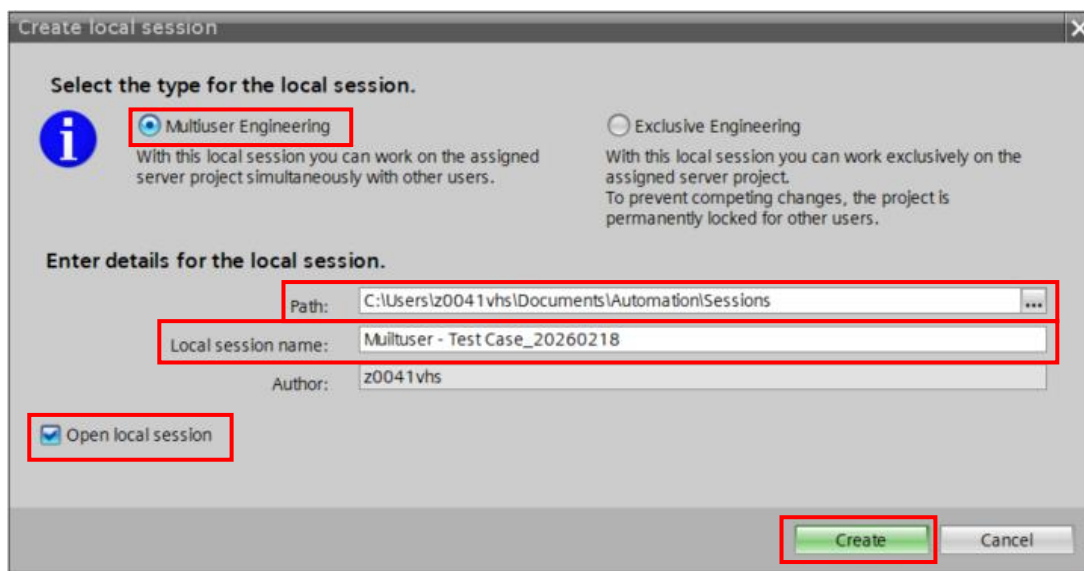


3. In the table below, you will see the currently hosted projects. Select the arrow next to the project **“Multiuser – Test Case”** to expand out the selections.
4. Double click the text **“Create New Local Session”**.



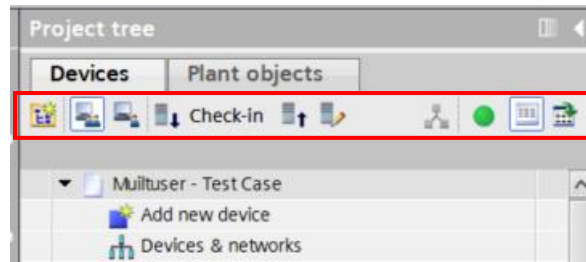
5. The **“Create Local Session”** dialog box will appear; There are several options that can be chosen here:
 - a. **Multiuser Engineering** – Simultaneously working on a project with other users.
 - b. **Exclusive Engineering** – This allows you to work on the project and lock it from other users.
6. For this guide, select the **“Multiuser Engineering”** radio button.
7. Enter the **“Path”**, this is where you want the project to be stored on your local computer. Use the **“...”** for assistance in navigating to the required path.
8. Enter the **“Local session name”**, this is an alias to let you know what version it is.

**** Its good to use the date here so that there is a record of when you downloaded the project ****
9. Tick the **“Open Local Session”** box.
10. Click **“Create”**.
11. The project will be downloaded to the local session folder and be opened up in TIA Portal.



Navigating The TIA Portal Project Tree

When using a project from the TIA Project Server with Multiuser Engineering, some of the icons at the top of the TIA Portal project tree will be added. These are as per the below:



	Creates A New Group (Standard TIA Function)
	Toggles Multiuser Mode On/Off
	Toggles Exclusive Multiuser Mode On/Off
	Check In Marked Changes
	Refresh Local Session
	Toggle Open/Close Server Project View
	Status Of Multiuser Commissioning Mode
	Status Of TIA Project Server
	Toggle Column Header
	Toggle Min/Max Overview

Toggling Multiuser Mode

TO BE COMPLETED

Toggling Exclusive Multiuser Mode

Toggling **“Exclusive Multiuser Mode”** allows a single user to take control/check out the complete project. This will stop others from accessing the project for editing.

Checking In Marked Changes

When changes

Refreshing Local Session

When using a project from the TIA Project Server with Multiuser Engineering, some of the icons at the top of the TIA Portal project tree will be added. These are as per the below:

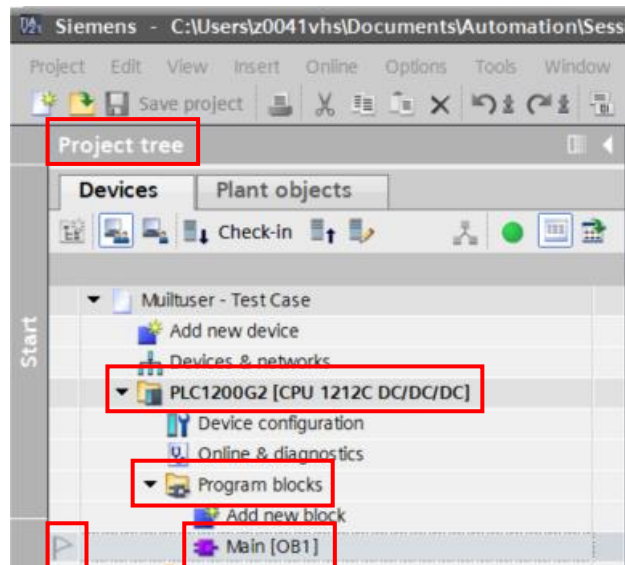
Marking And Unmarking Object And Flagging System

When a change is made in the local session, a flagging system is used to indicate the status of the item. The table below shows the meaning of each flag:

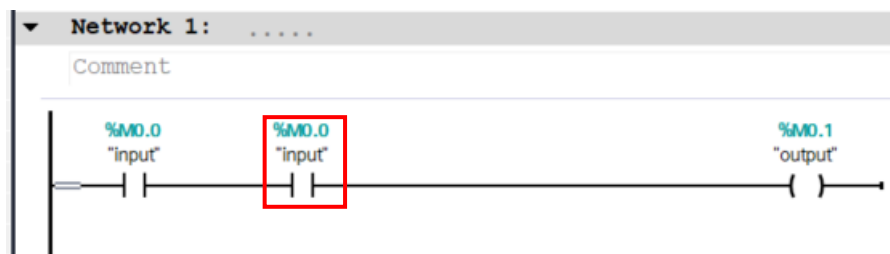
	Marked for check in by current session
	Marked by another session
	Not marked for check in
	Any of the above icons with the circled arrows means it need updating from the server

For example, before editing a function block the **“Grey”** flag will be shown. Once a change is made it will change to **“Blue”**, with a **“Yellow”** flag being displayed on any other clients with the ecosystem. To try this, follow the instructions below:

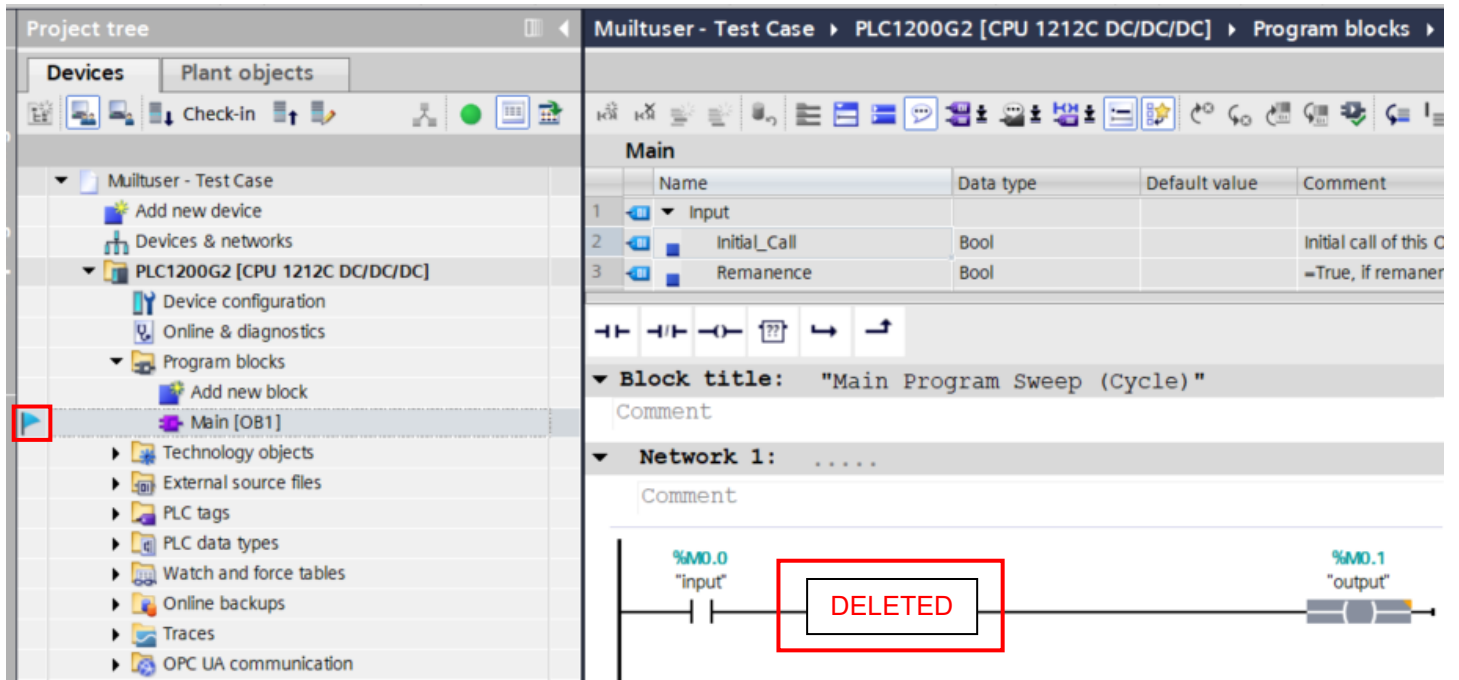
1. On **“engineering machine 1”**, navigate to **“Project Tree → PLC1200G2 → Program Blocks”** and double click **“Main [OB1]”**.
2. You will notice that next to **“Main [OB1]”** there will be a **“Grey”** flag indicating that it is not marked for check in.



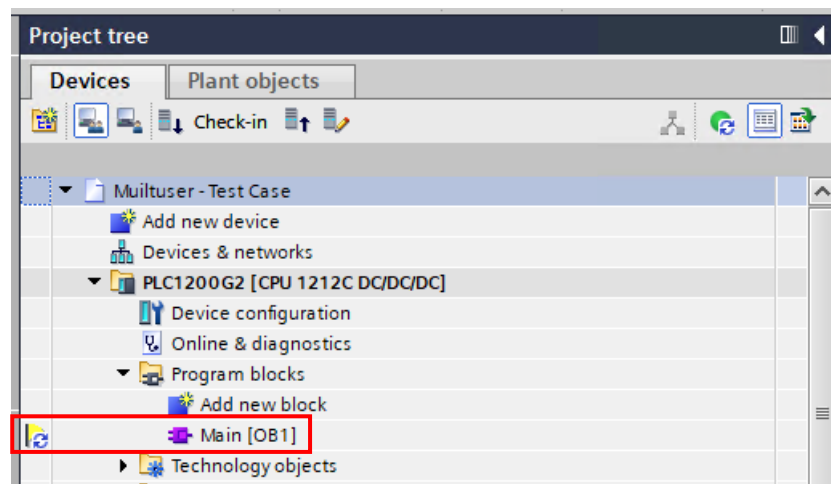
3. In the working area of TIA Portal, the internal coding of **“Main [OB1]”** will appear, you will notice that it has one line/rung of code. We want to make an edit here, delete the second instance of **“input %M0.0”**.



- Once deleted, you will notice that the flag in the project tree next to **"Main [OB1]"** will change to **"Blue"** meaning that it is marked for check in.



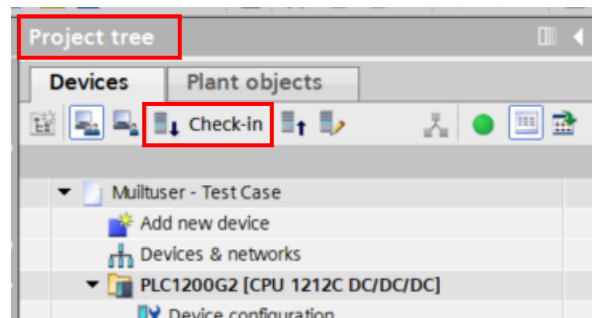
- If we now change to **"engineering machine 2"**, we should see that the flag next to **"Main[OB1]"** should be **"Yellow"** meaning the function block is marked by another user in another session. It also has the circled arrow icon which means it needs to be updated from the server.



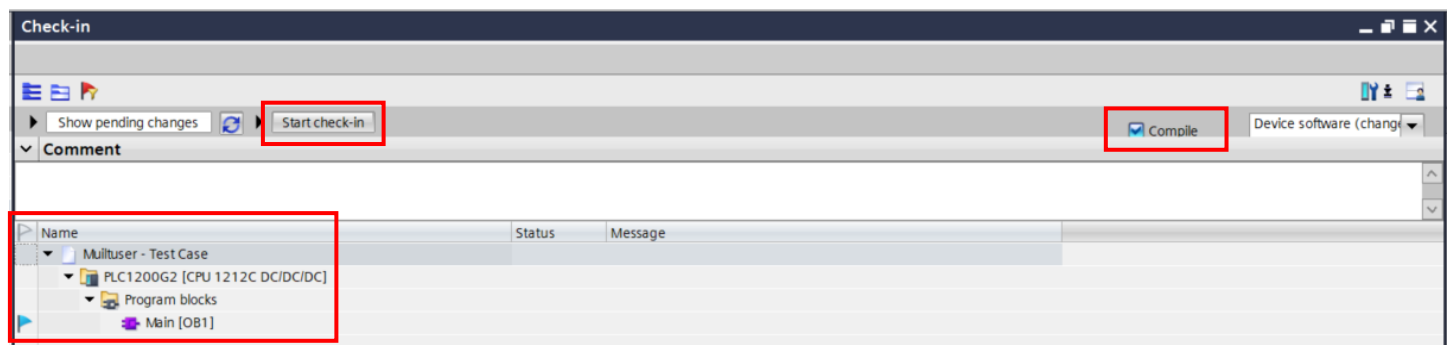
Checking In An Edited/Marked Object

Once the change is made and the object is marked for check in (**Blue Flag**), the engineer must then check the object back into the server. This ensures that the updates made can be shared amongst all the engineers collaborating on the same project. To check in an edited/updated object follow the below steps:

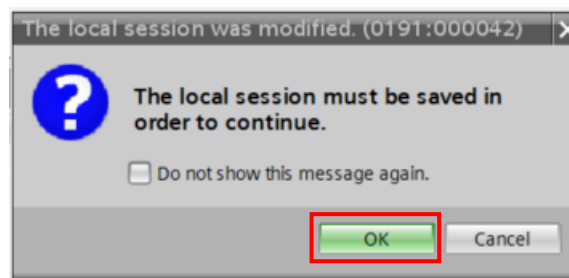
1. Navigate to the **"Project Tree"** and select the **"Check In"** button.



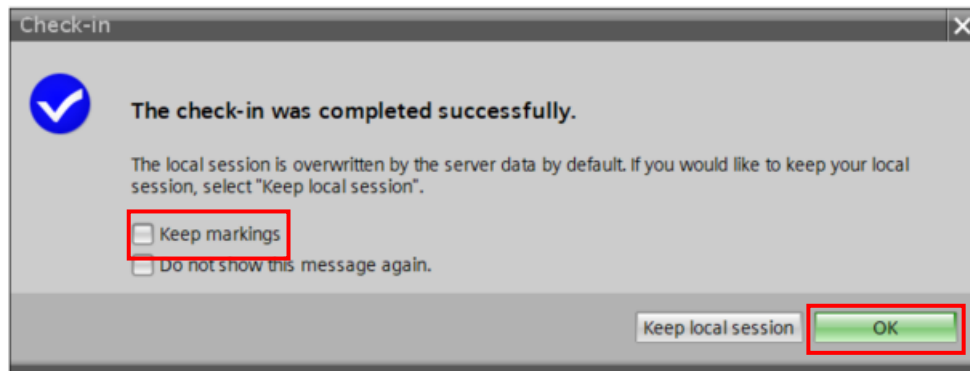
2. In the working area of TIA Portal, the **"Check In"** window will appear. It shows a summary of the items that are **"marked for check in"** which you can quickly navigate through using the expansion arrows on the left-hand side.
3. There is a **"Compile"** tick box in the right-hand corner of the window which allows you to compile any edits before they get sent back to server. This tick box is ticked by default, and it is recommended that a compile takes place before checking back in to ensure that another engineer does not receive any uncompiled work.
4. If the summary looks correct, click the **"Start Check In"** button in the top toolbar.



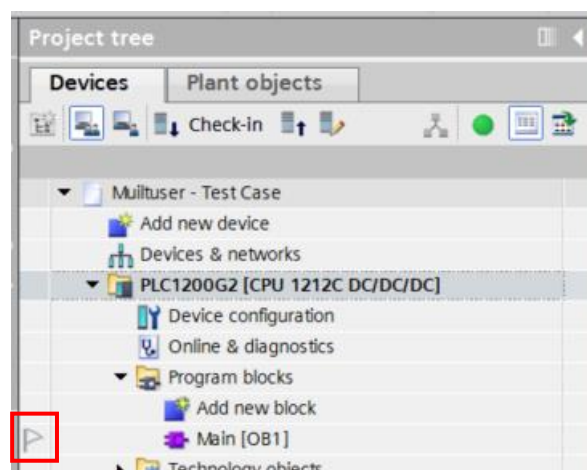
5. You might be prompted to **"Save"** the local session before you continue to check in the edits. Click **"OK"** to accept saving the local session.



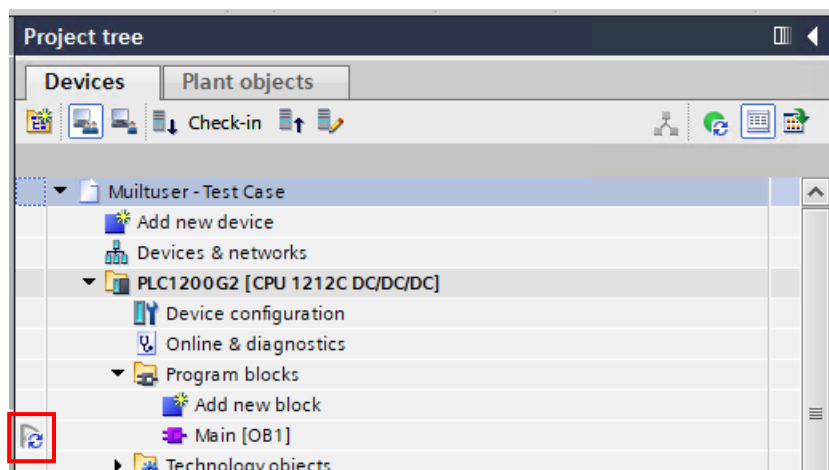
6. You will be prompted asking if you would like to overwrite the local session with the server data. What this means is that if other edits have been made they can be pulled down from the server at the time of synchronisation.
 - a. Selecting **"OK"** will download these changes to the local session.
 - b. Selecting **"Keep Local Session"** will not synchronise the changes with the server.
7. There is also a tick box called **"Keep Markings"** this means that you want to send the changes to the server, but you want to keep the other sessions informed that you are still working on the specific objects. This is good to use if you haven't quite finished but want to transfer the current changes to the server for backup reasons.
8. For this guide we will select **"OK"** and leave the **"Keep Markings"** unticked.



9. You will notice that the flag next to the **"Main [OB1]"** will now change to **"Grey without the circled arrows"** meaning that the item is up to date with the server and no changes have been made.



10. If we now change to **"engineering machine 2"**, we should see that the **"Main [OB1]"** flag has changed to **"Grey with circled arrows"** meaning that the function block is no longer being changed by another engineer but needs to be updated from the server. Updating the project from the project server is shown in the next chapter.



Abbreviations/Definitions

QDN – Qualified Domain Name

URL - Uniform Resource Locator

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